République Tunisienne
Ministère de l'Enseignement Supérieur
et de la Recherche Scientifique
Université de Tunis El Manar
Faculté des Sciences de Tunis
Département des Sciences de l'informatique

END-OF-STUDY PROJECT REPORT

Submitted in view of obtaining the national
diploma of Computer Engineering
Specialization: Embedded Systems and Internet of Things

By

Mdaissi Mouhib

Smart Recruitment System with Semantic AI Matching

Host Organization : Tunisie Telecom

President of jury : Mr Abbes haithem
Rapporteur : Mme Dammak Hajer
Supervised by : Mr Naouai Mohamed
Managed by : Mme Aissa Ahlem

Date de soutenance : 10/06/25

IOT - 3225022

Année Universitaire : 2024-2025

I authorize the student to submit their internship report for the purpose of a **final presentation of the project** .

Mme Aissa Ahlem

Signature and stamp

I authorize the student to submit their internship report for the purpose of a **final project presentation** ..

Academic Supervisor , Mr Naouai Mohamed

Signature and stamp

Acknowledgments

Contents

List of Figures

List of Tables

General Introduction

In a constantly evolving digital world, technology is reshaping the way individuals and organizations interact, and recruitment is no exception. Traditional hiring methods often suffer from inefficiencies, lack of personalization, and limited reach. These issues affect both job seekers and employers who aim to find the best fit as efficiently as possible.

In conventional recruitment systems, Human Resources personnel are required to manually sift through large volumes of CVs, assess each candidate's qualifications, and match them to job requirements a process that is often time consuming and inconsistent. Additionally, the lack of structured candidate data and intelligent filtering often leads to suboptimal hiring decisions and poor user experiences for applicants.

To address these limitations, we have developed a Smart Recruitment Web Platform that transforms the hiring process for both employers and job seekers. The system allows job seekers to register, build a profile, upload a CV, and apply to available positions. Employers, on the other hand, can publish job offers, manage incoming applications, and view AI-generated ranking scores to aid in candidate evaluation.

A standout feature of the platform is its integration with an AI-powered resume parser and matching system. This model automatically extracts relevant information from candidate CVs, compares them with job descriptions, and generates a CV matching score. These scores are displayed to recruiters to help them quickly identify the most qualified candidates, thereby optimizing the decision-making process and reducing time-to-hire. As a leading telecommunications company, "Tunisie Telecom" handles a large number of recruitment processes across various departments and roles. However, its current methods rely heavily on email applications, manual screening, and non-standardized candidate evaluations, which often result in delays, inconsistencies, and increased workload for HR staff. In this context, the need for an intelligent, centralized, and automated recruitment solution becomes evident. By adopting such a system, Tunisie Telecom can streamline its hiring process, improve the accuracy of candidate selection, and offer a more modern and efficient experience for both recruiters and applicants. This aligns with the company's ongoing efforts toward digital transformation and operational excellence. This report provides a complete overview of the project, structured into distinct phases. After introducing the general context and reviewing existing recruitment solutions in the "State of the Art," the report presents three major releases: the first focuses on user authentication and profile setup, the second on job offer management and candidate applications, and the third on the integration of an AI-based resume matching system. The report concludes with a general reflection on the system's outcomes and future perspectives.

Chapter 1

General Context

1.1 Introduction

This chapter outlines the general context of the project. It begins with a brief presentation of the host organization, followed by an overview of the project itself, including its purpose and objectives. The chapter concludes with a description of the development methodology adopted during the implementation phase.

1.2 Presentation of the Host Organization

1.2.1 Corporate Name and History:

Tunisie Telecom is the national and historical telecommunications operator in Tunisia. It was established in 1995 following the separation of postal and telecommunications services and has since grown to become the leading provider of telecom services in the country. It offers a broad range of services including fixed-line telephony, mobile communications, broadband internet (ADSL, fiber optics), cloud solutions, and enterprise services. [1]

Figure 1.1: TT Logo
[1]

1.2.2 Fields of Activity :

Tunisie Telecom is the incumbent telecommunications operator in Tunisia, providing a comprehensive range of services across the telecommunications sector. Its primary activities encompass:

- **Fixed-line and Mobile Telephony**: Offering voice communication services to both individual and corporate clients.
- **Internet Services**: Providing broadband internet access through technologies such as ADSL, VDSL, and fiber-optic connections.
- **Enterprise Solutions**: Delivering tailored communication solutions for businesses, including cloud computing and data services.

As of recent reports, Tunisie Telecom serves over six million subscribers in both fixed and mobile telephony within Tunisia and internationally. The company plays a pivotal role in enhancing the country's telecommunications infrastructure and expanding internet penetration rates. [1]

1.2.3 Organizational Structure of the Host Organization

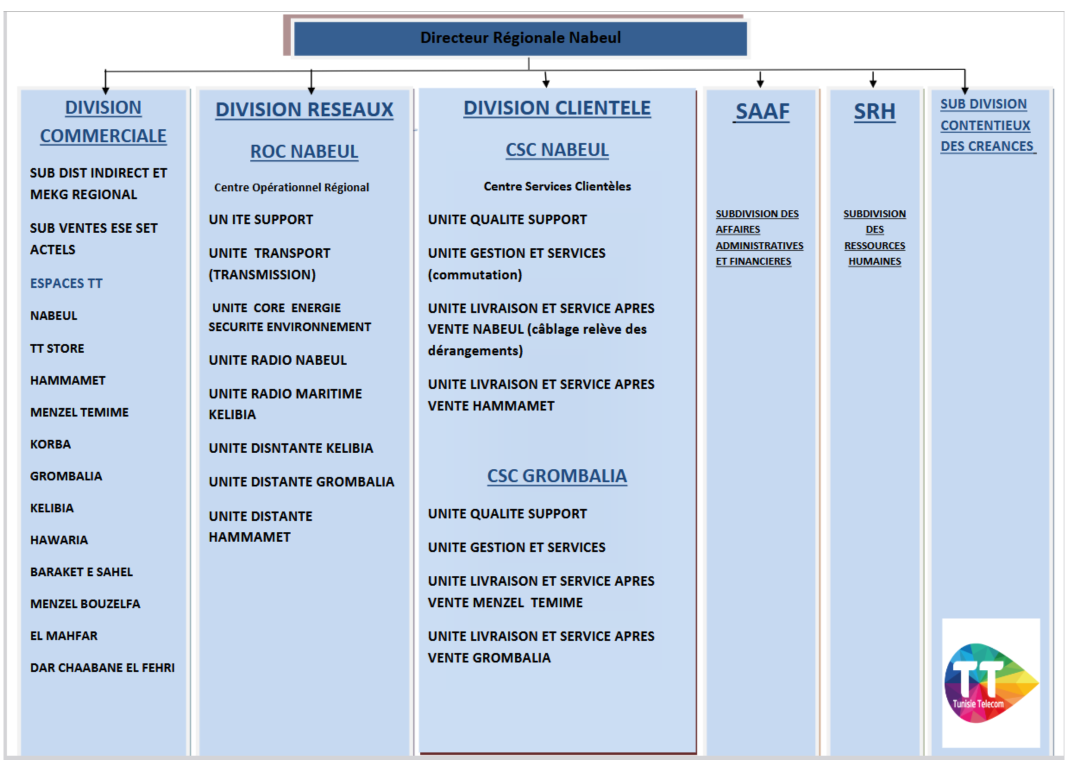

Figure 1.2: company structure

1.3 Presentation of the project

1.3.1 Problematic :

In today's digital age, the recruitment process remains largely dependent on manual operations, leading to inefficiencies, delays, and poor user experience. Human Resources teams are often overwhelmed with a high volume of applications and must review each resume individually without the support of intelligent tools. At the same time, candidates frequently face unclear application procedures, lack of feedback, and limited transparency in hiring decisions.

This reality highlights the necessity of developing a new recruitment solution the very foundation of this project which aims to address these limitations through a smart AI-powered web platform

1.3.2 Project Presentation

This project was developed as part of Final Year Project (FYP) required for obtaining the National diploma in Computer Engineering , specializing in Embedded Systems and IoT . It consists of designing and implementing a Smart Recruitment Platform that optimizes the hiring process for both employers and job seekers through automation, digitalization, and artificial intelligence.

The platform aims to address common limitations found in traditional recruitment processes by offering a complete and intelligent digital solution with the following key goals:

- **Simplification of the Recruitment Process**

  The main objective of the platform is to simplify the recruitment workflow for companies by providing a clear, intuitive interface where job offers can be published, applications tracked, and candidates evaluated in real-time.

- **Optimization of the Candidate Experience**

  From the job seeker's side, the platform focuses on user experience allowing fast profile creation, easy job searching, and transparent follow-up of applications.

- **Dedicated Management Tools for Recruiters**

  Employers have access to a full dashboard to manage their listings, review candidates, update application statuses, and automatically send feedback to applicants via email.

- **AI-Powered Resume Analysis**

  The system incorporates a deep learning module based on **BERT** (Bidirectional Encoder Representations from Transformers), capable of parsing resumes, extracting key information, and semantically comparing it against job descriptions. This process results in a **matching score** for each candidate, enabling recruiters to make more informed and efficient selection decisions.

1.3.3 Study of the Existing Systems

Before beginning the development of this platform, it was essential to analyze existing recruitment systems in order to identify their strengths, limitations, and opportunities for improvement.

Recruitment System at Tunisie Telecom

Tunisie Telecom currently uses a traditional recruitment process that relies on manual handling and fragmented communication .

Job Posting : Tunisie Telecom publishes job offers through multiple scattered channels including their official website, LinkedIn, social media , and job portals .

Application Submission: Candidates typically apply by sending their CVs and motivation letters via email or through basic external forms, without a structured or interactive application system.

Resume Screening: The screening and shortlisting process is manual, requiring HR staff to review each CV individually, which increases workload and slows down response times.

Candidate Communication: There is no automated notification system. Candidates are rarely updated on their status unless selected, leading to frustration and lack of transparency.

Tracking and Monitoring: Recruiters have no unified dashboard to monitor job openings, candidate
statuses, or application history everything is managed manually or via spreadsheets.

Rules Enforcement: Business rules (e.g., limiting candidates to one application) are not enforced, which can lead to duplicated or inconsistent applications.

Overall, the recruitment process at Tunisie Telecom is functional but outdated. It lacks automation, transparency, and speed which creates significant administrative burdens and affects both recruiter efficiency and candidate satisfaction.

Review of Existing Web-Based Recruitment Solutions

Several web-based platforms have been developed over the years to facilitate recruitment and connect employers with job seekers. Below is an overview of some of the most widely used platforms in the global and regional job market:

LinkedIn: is a professional networking platform that also offers powerful recruitment tools. Companies can post job offers, search for candidates using advanced filters, and interact with applicants directly through the platform. Users can build comprehensive profiles that serve as digital résumés, which are visible to recruiters based on their search criteria.[2]

Figure 1.3: Linkedin - Logo

SmartRecruiters: is a modern, enterprise-grade recruitment software that provides end-to-end talent acquisition capabilities. It supports job posting, candidate sourcing, applicant tracking, collaborative hiring, and recruitment analytics. The platform also integrates with external job boards and AI tools for resume screening and matching . [3]

Figure 1.4: SmaetRecruiters - Logo

Glassador : is a comprehensive employment platform that offers job listings, company reviews, salary information, and interview insights. Employers can enhance their brand presence by creating detailed company profiles, posting job openings, and engaging with a pool of informed candidates. Job seekers benefit from access to a wealth of information, including employee reviews, salary data, and interview experiences, enabling them to make more informed application decisions.[4]

Figure 1.5: Glassador - Logo

Criteria	**LinkedIn**	**SmartRecruiters**	**Glassdoor**
Type	Professional network with integrated recruitment tools	Cloud-based enterprise recruitment platform (full ATS)	Job site combined with company reviews and salary insights
Cost	Free options and paid recruiter plans	SaaS model with custom enterprise pricing	Free job postings with sponsored listing options
Ease of Use	User-friendly interface; advanced features may require onboarding	Professional, complete interface; some features require training	Simple, user-friendly for candidates and employers
Job Posting	Quick posting with filters and profile search	Advanced multi-channel job posting with campaign tracking	Job postings shown on company review pages
Smart Matching	Keyword-based; AI profile suggestions	AI-driven resume parsing and job-description matching score	Not focused on AI matching; more on exposure and branding
Candidate Management	Recruiter dashboard with ATS integrations	Built-in ATS and full recruitment pipeline tools	Basic application alerts; limited tracking features
Inconvenients	• Limited AI in free version • Premium plans can be costly • More suited for professionals	• Complex for small teams • Paid-only access • Requires training/onboarding	• Not a full ATS • Emphasis on branding, less on automation • Limited recruiter tools
Best For	Recruiting professionals, freelancers, and B2B talent	Medium to large enterprises with complex HR workflows	Recruiters focusing on employer branding and public listings

Table 1.1: Comparative Analysis of Web-Based Recruitment Platforms

Critical Analysis :

The comparative table illustrates the key features and limitations of three prominent recruitment platforms: LinkedIn, SmartRecruiters, and Glassdoor. While each of these tools plays a significant role in the digital hiring ecosystem, the analysis also reveals clear gaps that justify the development of a more comprehensive and intelligent platform. A major observation is that most existing solutions tend to specialize in one specific aspect of the recruitment process networking, applicant tracking, or employer branding but none provide a unified, lightweight, and intelligent system tailored to both small employers and job seekers in a streamlined manner. For instance, LinkedIn focuses primarily on professional networking, SmartRecruiters on applicant tracking, and Glassdoor on employer branding.

1.4 Proposed Solution

In response to the challenges identified in traditional recruitment processes such as manual screening, lack of communication, and inefficiencies this project introduces the **TT Recruit System**, a smart, web-based recruitment platform. The system integrates automation, role-based access, and intelligent decision-making to streamline the entire hiring process. It not only facilitates job posting and application tracking but also enhances efficiency, transparency, and fairness through AI-driven candidate matching and real-time email notifications. The table below presents a detailed comparison between the existing system used by Tunisie Telecom and the proposed solution.

Recruitment Aspect	**Tunisie Telecom (Traditional System)**	**TT Recruit System (Proposed Solution)**
Job Posting	Published via multiple channels (website, LinkedIn, email)	Centralized job creation and publishing via recruiter dashboard
Application Submission	Through email or basic external forms	Structured online application with resume upload
Resume Screening	Manual CV review by HR staff	AI-powered automatic CV parsing and job matching
Shortlisting Process	Based on manual screening	Intelligent candidate ranking and filtering
Application Tracking	No unified tracking; status often unknown	Real-time application tracking for recruiters and job seekers
Candidate Communication	Inconsistent or absent communication	Automated email updates throughout the recruitment process

Recruitment Aspect	**Tunisie Telecom (Traditional System)**	**TT Recruit System (Proposed Solution)**
Email Notification System	No automated system; all communication is manual	Email notifications triggered for: new job posts, status updates, interviews, decisions
Rules Enforcement	No control over duplicate or multiple applications	Enforces one active application per job seeker
Time to Hire	Long due to manual processing and lack of automation	Significantly reduced through automation and centralized control
Candidate Experience	Frustrating, unclear, and passive	Transparent, interactive, and well-informed journey
Justification & Impact	Traditional system struggles with volume, coordination, and communication	Modern, scalable, and intelligent system that improves efficiency and satisfaction

Table 1.2: Comparison Between The Traditional Syste and TT Recruit System

1.5 Methodology

A methodology in software development defines the process and structure used to organize tasks, manage progress, and deliver a final product. It serves as a guide for how the project is planned, executed, and evaluated. Choosing the right methodology is essential for ensuring efficiency, adaptability, and collaboration throughout the development lifecycle. Depending on the project's nature and constraints, different approaches such as Waterfall, Agile, or Kanban may be adopted. In the case of this project, a comparison of these methodologies was conducted to determine the most appropriate framework for building a modular, scalable, and evolving web-based application.

1.5.1 Comparison of Project Management Methodologies

Criterion	Waterfall Method	Agile Method (Scrum)	Kanban Method
Approach	Sequential, step-by-step	Iterative and incremental	Continuous flow
Flexibility	Low	High	High
Change Management	Difficult to implement	Easy to integrate	Easy to integrate
Delivery	At the end of the project	At the end of each sprint (2–4 weeks)	Continuous
Client Involvement	At the beginning and end only	Continuous	Continuous
Planning	Fixed and predefined	Adaptive and evolving	Adaptive
Visibility	Low before the end of the project	High thanks to regular meetings and sprints	High due to visual tracking
Risk	High due to long phases without feedback	Reduced through short iterations	Reduced through continuous adjustments
Collaboration	Low, mostly internal	High, self-organizing team	High, self-organizing team
Retrospective and Continuous Improvement	Rare, usually at the end of the project	Integrated after each sprint	Ongoing

Table 1.3: Comparison of Project Management Methodologies

1.5.2 Agile :

Agile is a project management methodology based on iterative development, where requirements and solutions evolve through collaboration between cross-functional teams and end-users. It emphasizes flexibility, continuous delivery, and customer feedback. [5]

Figure 1.6: Agile Methodology

1.5.3 Methodology Adapted : Scrum

Scrum is a lightweight framework designed to help teams collaborate on complex products iteratively and incrementally to deliver high value. It is built on the principles of empiricism and lean thinking.[6]

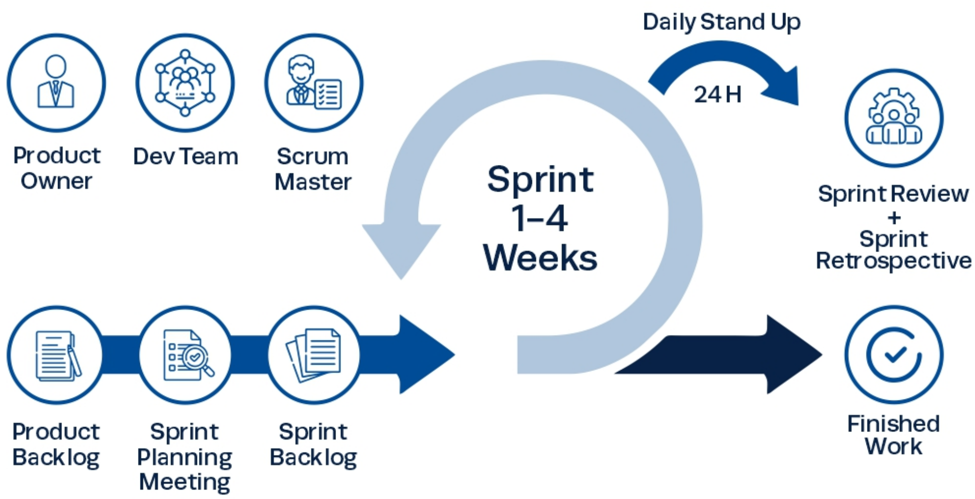

Figure 1.7: Scrum Process Diagram

To ensure the smooth progress of the project and structure its different phases effectively, I adopted Scrum as a project management methodology. Although typically used by teams, Scrum principles can also be applied individually to promote organization, iteration, and continuous improvement. The method is based on short, time-boxed iterations called sprints, which generally last between one and four weeks. In this project, I followed two-week sprints to define goals, implement functionalities, and assess progress. At the end of each sprint, I reviewed the completed work and adjusted the next steps accordingly. Several key elements define the Scrum framework:

- The **Product Backlog** is a list of prioritized requirements and features, initially defined and then continuously updated throughout the project based on evolving needs.
- The **Sprint Backlog** is a subset of items from the product backlog selected for implementation during a given sprint, based on a defined objective.
- A **User Story** refers to a short, client centered description of a functionality to be developed.

- The **Daily Scrum**, is traditionally a brief daily meeting for team synchronization. In my context, I replaced it with a daily self-review to monitor progress and adjust priorities.

This structure helped maintain project coherence, ensure continuous progress, and support iterative improvements despite the absence of a formal team.

Scrum Roles :

- Product Owner : manages the product backlog and defines priorities
- Scrum Master: ensures the process is followed and removes obstacles.
- Development Team: cross-functional team that builds the product.

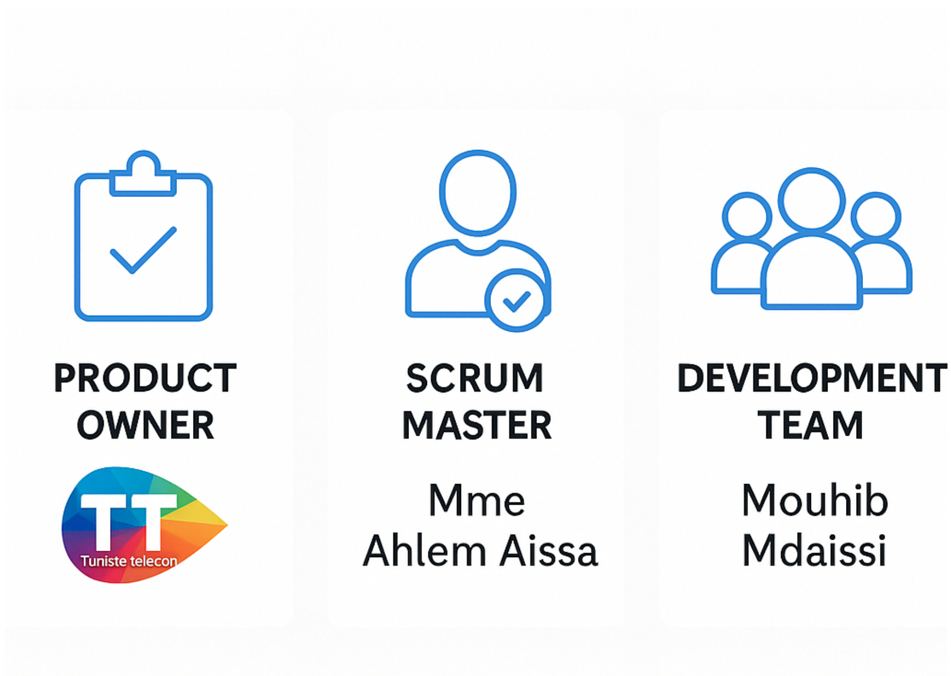

Figure 1.8: Role distribution

1.6 Conclusion

This chapter introduced the project's context, objectives, and the motivation behind developing the TT Recruit System. The next chapter will detail the system's requirements, architecture, technologies, and the Scrum-based project management approach.

Chapter 2

State of the Art

2.1 Introduction

In this chapter, we explore all the key elements needed to build the TT Recruit System. We start by identifying the main users of the system and describing what the system should do (its functional and non-functional requirements). Then, we present a general view of the system using diagrams and modeling tools to help visualize how it works. After that, we explain the technologies used in the project, especially the MERN stack, and how the different parts of the system are connected. Finally, we describe how the project was managed using the Scrum method, including how we planned the work with a product backlog and sprints.

2.2 Identification of Actors

Actors are the entities that interact with the system in order to use its services or to manage its functionality. For this project, two main actors were identified:

- **Job Seeker:** The job seeker is a primary user of the platform. This actor can register and log in to the system, complete a personal profile including uploading a resume (PDF format), and view available job offers posted by the company. The job seeker can apply to only one job at a time, as the platform is designed for use by a single company. Once the application is submitted, the job seeker can track his application status (accepted, rejected, or pending) through their dashboard.
- **Recruiter:** The recruiter is the person in the company (HR) responsible for publishing job offers. This actor can create and delete job postings and consult the list of candidates who have applied to each job. The recruiter can accept or reject candidates and view AI-based matching scores based on resume content.

2.3 Requirements Specifications

2.3.1 Functional Requirements

The functional requirements define what the system should be able to do from a user perspective. These are described in terms of actions that the jobseeker and the recruiter can perform on the platform.

JobSeeker: An individual looking for job opportunities who can apply to job offers via the platform.

Recruiter: A company representative responsible for posting job offers and managing applicants.

For the JobSeeker

- Register and log in securely with Jobseeker user role
- Complete personal profile and upload a resume
- View the list of available job offers
- Apply for selected job offers
- Track application status (pending, accepted, rejected)
- Receive OTP by email to password reset
- Receive an email notification when a new job is posted
- Receive an email notification upon acceptance or rejection of their application

For the Recruiter

- Register and log in securely with Recruiter user role
- Receive OTP by email to verify email
- Receive OTP by email to password reset
- Create and delete job postings
- View list of applicants per job offer
- Accept or reject applications
- Access AI-based resume matching scores

2.3.2 Non-Functional Requirements

Non-functional requirements define how the system should behave in terms of performance, security, and usability. These are critical to ensure the robustness and quality of the solution.

- **Performance:** The system must ensure a quick response time during browsing, job application, and dashboard interactions. Resume parsing and score generation should be processed efficiently.
- **Security:** All sensitive operations such as login, application submission, and file upload must be secure. JWT-based authentication and encrypted cookie storage are used to protect user sessions. Only PDF files are allowed for resumes, and images for profile photos.
- **Scalability:** The application should be capable of supporting future expansion in terms of user base and features, especially in handling large volumes of job offers and applications.
- **Usability:** The interfaces must be intuitive and user-friendly, particularly for the job seekers who interact with forms and dashboards frequently.

2.4 Global System View

2.4.1 Modeling Language

The Unified Modeling Language (UML) is a graphical language for visualizing, specifying, constructing, and documenting the artifacts of a software-intensive system. The UML gives you a standard way to write a system's blueprints, covering conceptual things, such as business processes and system functions, as well as concrete things, such as classes written in a specific programming language, database schemas, and reusable software components. [7]

Figure 2.1: StarUML - Logo

2.4.2 UseCase Diagram

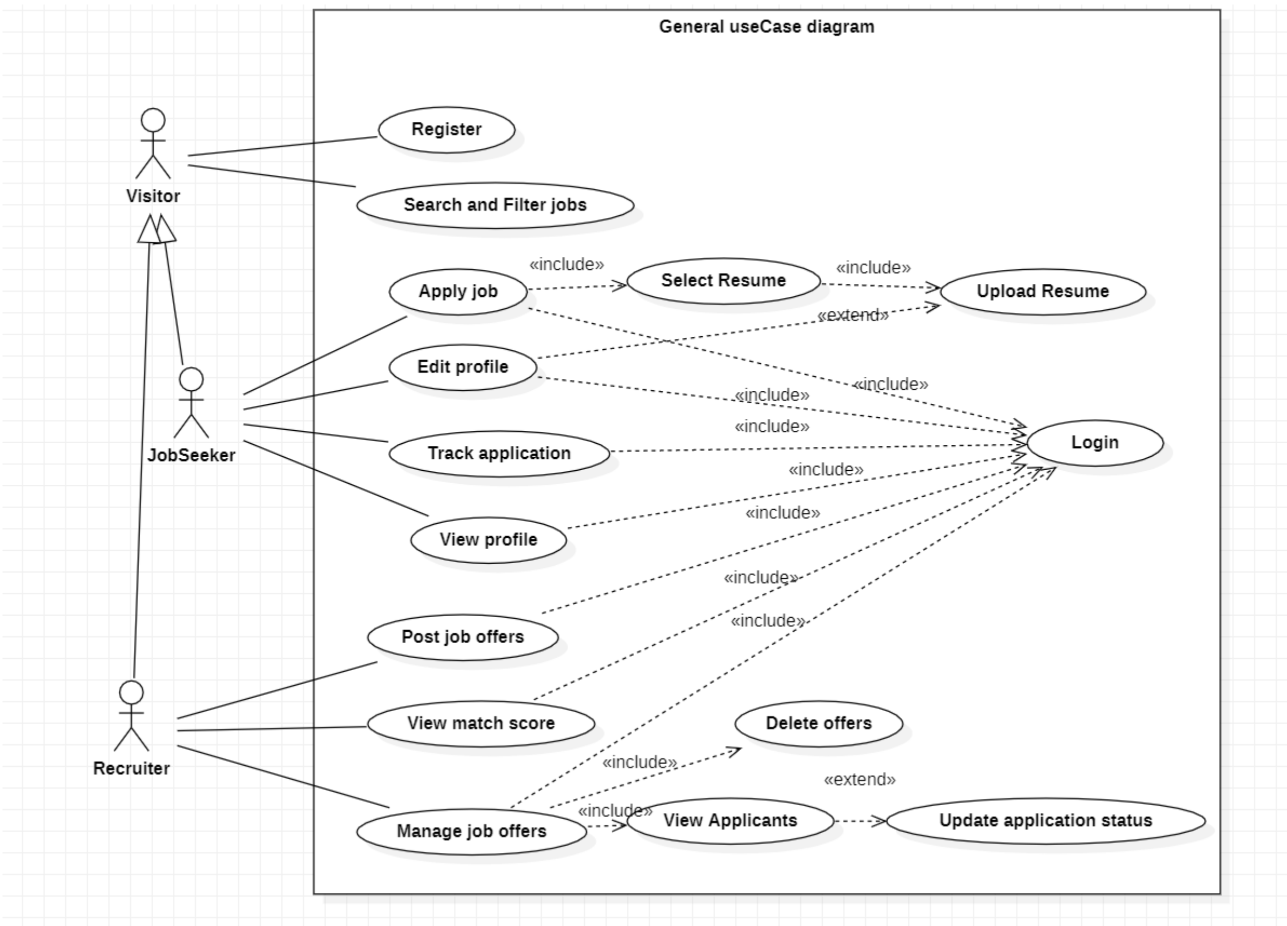

Figure 2.2: useCase Diagram

Figure 2.2 is the general use case diagram that illustrates the main functionalities of the recruitment system and how different users interact with it. There are three actors: *Visitor*, *JobSeeker*, and *Recruiter*. Visitors can register and view job listings, while job seekers, who inherit from visitors, can also apply for jobs and update their profiles. Recruiters have access to job management functionalities such as posting, deleting, and managing jobs, as well as viewing applicants, updating application statuses, and using the AI matching system. The *Login* use case is a required step for accessing most system features, as indicated by the include relationships. The diagram provides a clear view of the system's access control and role based functionalities.

2.4.3 Class Diagram :

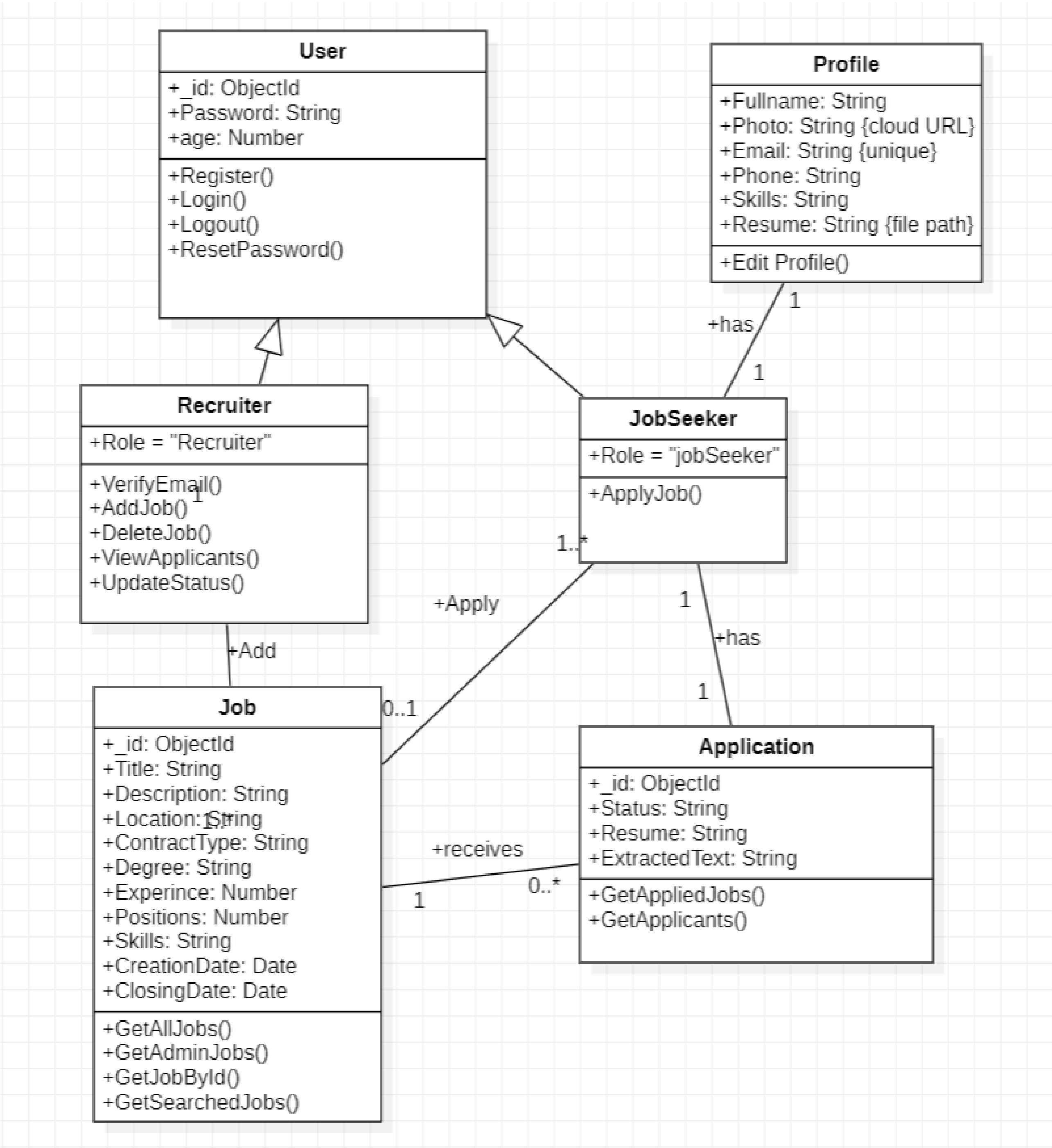

Figure 2.3: Class Diagram

Figure 2.3 is the class diagram that illustrates the core structure of the TT Recrut System. At the center of the system is the User class, which provides shared authentication and identity functionality. Each user has a one-to-one relationship with a Profile that contains personal details such as full name, email, phone, skills, profile photo, and resume. The system supports two specialized user types through inheritance: Recruiter and JobSeeker. Recruiters can add and delete job offers, view applicants, and update application statuses. JobSeekers can apply to available job offers. A job seeker may apply to only one job at a time, enforced through logic, though the application history is retained. Each `Job` is created by a recruiter and includes attributes such as title, location, contract type, experience level, and deadlines. Applications are associated with both a job and a job seeker, and include a resume, status, and optionally, extracted resume text used in AI matching. Overall, this diagram reflects the system's main entities, their attributes, and the operations relevant to each role, supporting both the recruitment and application workflows.

2.5 Software Architecture

2.5.1 MERN Stack Architecture

This project is built using the **MERN stack**, a widely adopted technology stack for full-stack JavaScript development. MERN is an acronym for: **MongoDB** , **Express.js** , **React.js** , **Node.js** .
The MERN stack offers a unified JavaScript environment for both client-side and server-side development. This architectural choice ensures high developer efficiency, seamless RESTful API integration, and support for scalable single-page applications (SPA).

2.5.2 Overall System Architecture

The TT Recruit System is composed of distinct yet interconnected components: a React-based frontend for user interaction, a Node.js/Express backend for core logic and data management, and a Flask-powered AI microservice for resume-job matching. These modules communicate through RESTful APIs, ensuring a clean separation of concerns and enabling smooth data flow across the system.

- **Frontend**: Implemented using React.js, this layer is responsible for handling all user interactions. It provides interfaces for user registration, authentication, profile creation, job browsing, and application submission. Communication with the backend is achieved via secure REST API calls. it includes two separate dashboard views tailored to each user role. The Recruiter dashboard provides tools for managing job listings and viewing applicant scores, while the Job Seeker dashboard allows users to search for jobs, apply, and track their application statuses.

- **Backend**: The core application logic resides here, built using Node.js and Express.js. It handles API routing, request validation, and access control through middleware such as role-based authentication and authorization mechanisms. The backend is also responsible for managing job postings, user profiles, and application records. MongoDB serves as the primary data store, maintaining collections for users, jobs, and applications, including metadata like timestamps, roles, and matching scores.
- **AI Matching Microservice**: The system's intelligence is encapsulated in a Python-based microservice built with Flask. This service receives resume text and job descriptions from the backend and uses a fine-tuned BERT (Bidirectional Encoder Representations from Transformers) model to calculate a semantic similarity score between them. This score quantifies how well a candidate's resume aligns with a job's requirements and is returned to the backend for storage and ranking.
- **Data Flow**: When a user applies for a job, the backend extracts the resume content and forwards it, along with the job description, to the AI service. The resulting match score is stored alongside the application data in MongoDB. Recruiters can then view a ranked list of applicants, sorted by their relevance to the job posting, thus enabling faster and more accurate shortlisting.

This modular architecture promotes separation of concerns, facilitates easier maintenance, and ensures that the AI logic is loosely coupled and independently scalable. It also supports future extensibility for instance, the AI service could be replaced or extended with a more advanced model without affecting the rest of the system.

2.6 Project Management Using Scrum

2.6.1 Product Backlog

Actor	Feature	User Story
Visitor	**Registration/Login**	As a visitor, I want to register and log in securely with my role (JobSeeker or Recruiter).
	Search and View Jobs	As a visitor, I want to browse and search available job offers.
JobSeeker	**Registration/Login**	As a JobSeeker, I want to register, log in securely and get directed to my dashboard.
	Complete Profile	As a JobSeeker, I want to complete and update my personal profile.
	Upload Resume	As a JobSeeker, I want to upload my resume in PDF format.
	Edit Profile	As a JobSeeker, I want to be able to edit my profile after registration.
	Browse Job Offers	As a JobSeeker, I want to browse available job offers posted by the company.
	Search / Filter Jobs	As a JobSeeker, I want to filter or search job offers by title or location.
	Submit Application	As a JobSeeker, I want to apply for an available job using my profile resume or upload a new one.
	Track Application Status	As a JobSeeker, I want to see if my application was accepted, rejected, or still pending.
	Receive Email (New Job)	As a JobSeeker, I want to receive an email when a new job is posted.
	Receive Email (Application Result)	As a JobSeeker, I want to receive an email when my application status is updated.
	Dashboard	As a JobSeeker, I want a dashboard showing my resume and application status.
Recruiter	**Registration/Login**	As a recruiter, I want to register using my company email so that I can access the system after verifying my email with an OTP.
	Post Job Offer	As a Recruiter, I want to create job offers.
	Delete Job Offer	As a Recruiter, I want to delete job offers.
	View Applicants	As a Recruiter, I want to view all candidates who applied to my job offers.
	View Resume	As a Recruiter, I want to preview the candidate resume.
	See Match Score	As a Recruiter, I want to see a match score for each applicant based on their resume and job offer.
	Accept / Reject Applications	As a Recruiter, I want to accept or reject applications and update their status.
	Dashboard	As a Recruiter, I want a dashboard showing all job offers and received applications.

Table 2.1: Product Backlog Organized by Actor

2.6.2 Sprints Planning

Release 1			
Sprint	**Dates**	**Sprint Title**	**Main Objectives**
Sprint 1	Feb 10 – Feb 24	**User Identification**	• Setup MongoDB User schema. • Implement JWT login and role-based access • Role selection during signup • Email verification for the recruiter • upload profile image during signup • store the image in cloud storage
Sprint 2	Feb 25 – Mar 10	**Setup The User Profile**	• Build candidate profile page • Upload and store resume (PDF) • Save resume to Disk storage • The user can update his profile

Table 2.2: Release 1

Release 2			
Sprint	**Dates**	**Sprint Title**	**Main Objectives**
Sprint 3	Mar 11 – Mar 24	**Employer Job Management**	• Setup MongoDB Job schema • Recruiter dashboard layout • Create and delete job offers • Display list of job offers posted by logged-in recruiter • Identify expired job offers • Email notification for new job postings
Sprint 4	Mar 25 – Apr 7	**Job Application Management and JobSeeker Interface**	• Setup MongoDB Application schema • Implement "Apply to Job" logic • Allow resume selection (use existing or upload new) • Job search and filter jobs logic • JobSeeker dashboard layout • Display application history to JobSeeker • View list of applicants per job (by Recruiter) • Update applications status • Email notification for updates on application status

Table 2.3: Release 2

Release 3			
Sprint	**Dates**	**Sprint Title**	**Main Objectives**
Sprint 6	Apr 22 – May 5	**AI-Based Resume Matching System**	• Integrate Python AI microservice (BERT) • Extract fields from resumes • Match resume vs job & generate score • Display score in recruiter dashboard

Table 2.4: Release 3

Before starting the implementation sprints, a 10-day period was dedicated to project setup and analysis. During this phase, I defined the project scope, selected the appropriate technologies (React, Node.js, Express, MongoDB, Tailwind), and set up the development environment. I also analyzed the functional requirements, identified the main users and features, and designed the overall architecture to ensure a clear and structured development process in the upcoming sprints.

2.7 Work environment :

2.7.1 Material environment :

During the development and testing phase of our application, we are using a personal computer with this specifications :

Processor	11th Gen Intel® Core™ i5-11400H @ 2.70GHz
SSD	477 GB
RAM	8 GB (3200 MHz)
Graphics Card	4 GB (Multiple GPUs)
OS	Windows 11 (x64)

Table 2.5: Machine specifications used during development

2.7.2 Software environment :

1. **Visual Studio Code (VScode):** Visual Studio Code is a lightweight but powerful source code editor developed by Microsoft. It supports debugging, embedded Git control, syntax highlighting and extensions for a wide range of programming languages and tools. [8]

Figure 2.4: VScode - Logo

2. **Postman :** An API testing tool used during development to validate and debug endpoints between the frontend and backend [9]

Figure 2.5: Postman - Logo

3. **Git :** Git is a free and open-source distributed version control system built to manage projects of all sizes with high speed and efficiency. [10]

Figure 2.6: Git - Logo

4. **Github :**GitHub is a platform that hosts projects managed with Git. It allows developers to publish their local repositories online and collaborate by contributing to publicly shared repositories maintained by others. [11]

Figure 2.7: Github - Logo

5. **MongoDB Compass :** Compass is a free, user-friendly interface that allows you to query, analyze, and optimize your MongoDB data interactively. [12]

Figure 2.8: MongoDB Compass - Logo

2.7.3 Technologies :

1. **React.js :** React is a JavaScript library developed by Meta (Facebook) for building dynamic user interfaces. It allows developers to create reusable UI components and efficiently update views based on state changes. [13]

Figure 2.9: Reactjs - Logo

2. **Javascript (js) :** JavaScript is a high-level, interpreted programming language primarily used to create interactive and dynamic content in web applications. It runs in the browser and enables features like form validation, animations, and asynchronous communication with servers. It is also widely used on the server side with environments like Node.js. [14]

Figure 2.10: JS - Logo

3. **Tailwind CSS :** Tailwind CSS is a utility-first CSS framework that enables developers to build custom user interfaces directly in their HTML by applying predefined classes, offering flexibility without writing custom CSS. [15]

Figure 2.11: Tailwind - Logo

4. **Node.js :** Node.js is an open-source, cross-platform runtime environment that allows developers to run JavaScript code on the server side.[16]

Figure 2.12: Nodejs - logo

5. **Express.js :** Express is a minimal and flexible Node.js web application framework that provides a robust set of features for building APIs and web applications. It simplifies routing, middleware integration, and server configuration. [17]

Figure 2.13: Express.js - Logo

6. **MongoDB :** MongoDB is a NoSQL, document-oriented database designed for scalability and flexibility. It stores data in BSON (binary JSON) format, enabling efficient storage and retrieval of complex and hierarchical data structures.[18]

Figure 2.14: MongoDB - Logo

7. **Nodemailer :** Nodemailer is a Node.js module that allows applications to send emails easily. [19]

Figure 2.15: Nodemailer - logo

2.8 Conclusion

In this chapter, we identified both the functional and non-functional requirements of the system. We also presented the product backlog, the global use case diagram, and the system's class diagram to illustrate the main components and interactions. Additionally, we defined the system architecture and outlined the work environment ,the planned sprints according to the Scrum methodology. These elements together provide a clear and structured foundation for development. In the next chapter, we will focus on implementing **Release 1** of the project

Chapter 3

Release 1 :User Authentication and Profile Setup

3.1 Introduction

This chapter covers the initial setup of the TT Recruit System, including secure user authentication and profile creation. It details the features developed in Sprint 1 and Sprint 2, such as login, registration, and profile management.

3.2 Sprint 1 : User identification

3.2.1 Sprint Backlog

Actor	Feature	User Story
User	Register	As a user, I want to be able to create an account
	Role Selection	As a user, I want to select my role (JobSeeker or Recruiter) during registration.
	Login	As a user, I want to log in securely with my credentials.
	Dashboard Redirection	As a user, I want to be redirected to the correct dashboard based on my role.
Recruiter	Email Verification	As a recruiter, I want to verify my company email using an OTP before completing registration.

Table 3.1: Sprint 1 Backlog

3.2.2 Functional Specification

Use Case Diagram:

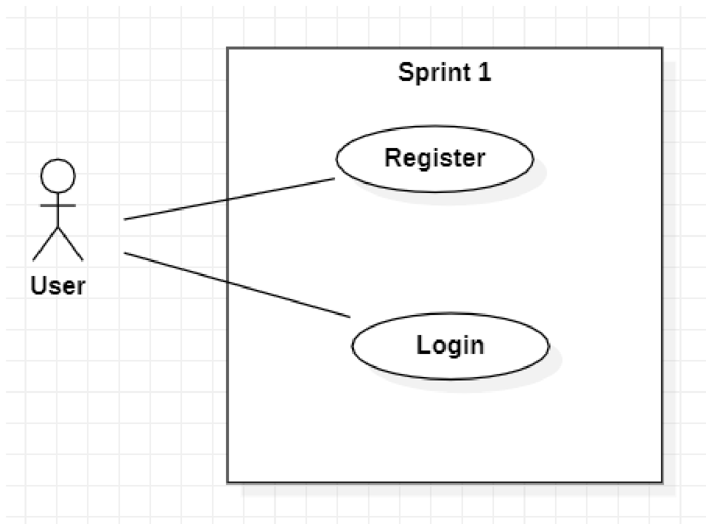

Figure 3.1: Sprint1 - Use case daigram

Description

- **UseCase : User – Register :**

This use case allows a new user to create an account on the platform. During registration, the user is required to select their role: either Employer or Job Seeker. If the user selects Job Seeker, they can proceed directly by filling in their personal details and completing the registration. However, if the user selects Employer, the system checks whether the email address used belongs to a valid company domain. The registration process does not complete immediately; instead, an OTP is sent to the provided company email. The user must enter the OTP to verify their email. Only after successful verification is the account created. The system then stores the user's role and redirects them to the corresponding dashboard.

- **UseCase : User – Login :**

This use case allows a registered user (either a recruiter or a job seeker) to log into the system. The user provides their email and password, and the system verifies the credentials. If the login is successful, the user is redirected to their respective dashboard based on their assigned role. If the credentials are incorrect, an error message is displayed prompting the user to retry. No role selection is required during login, as the system automatically recognizes the user's role after authentication.

Conception

- **Sequence Diagram of the User Registration Process**

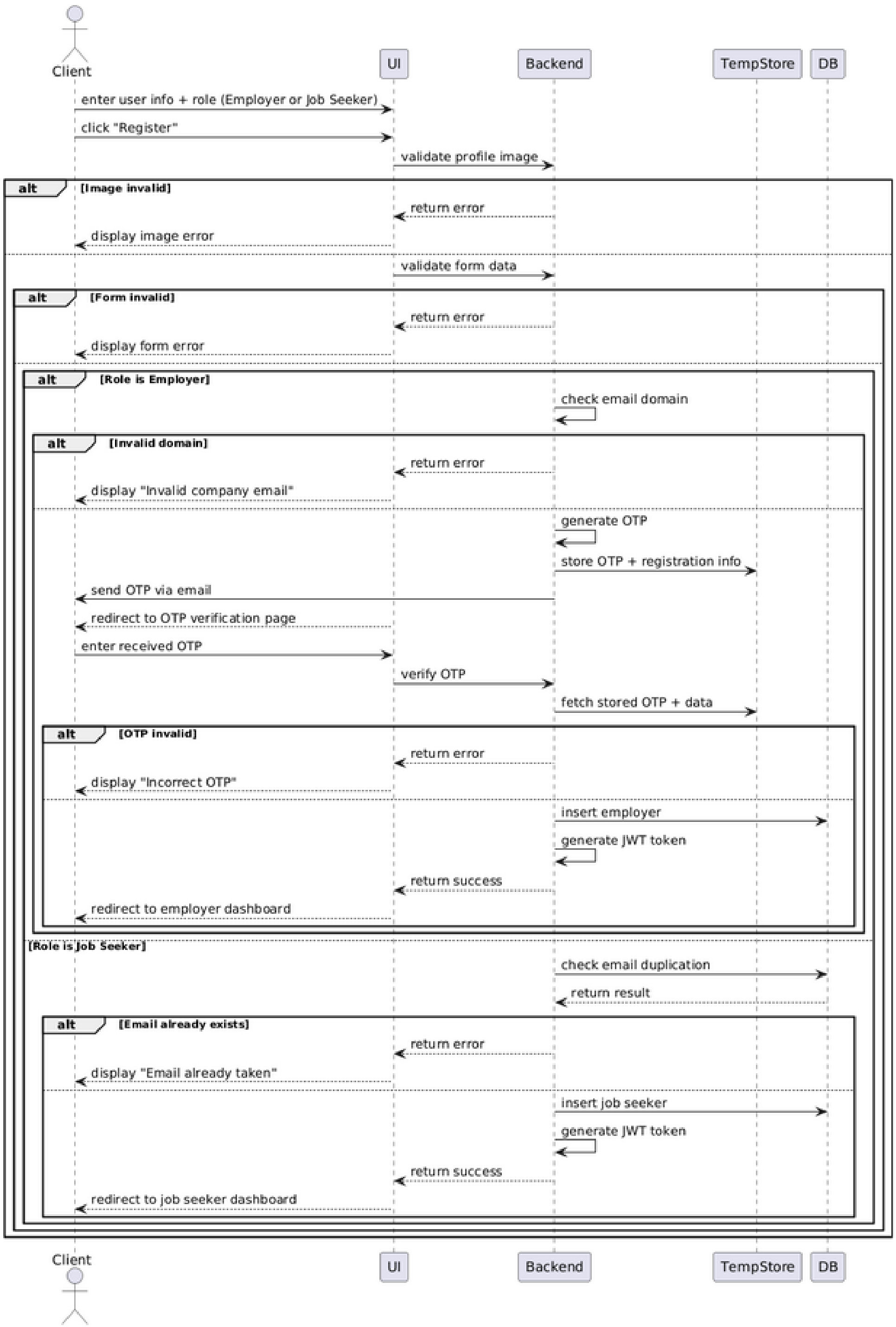

Figure 3.2: Sequence Diagram - Register

- **Sequence Diagram of the User Login Process**

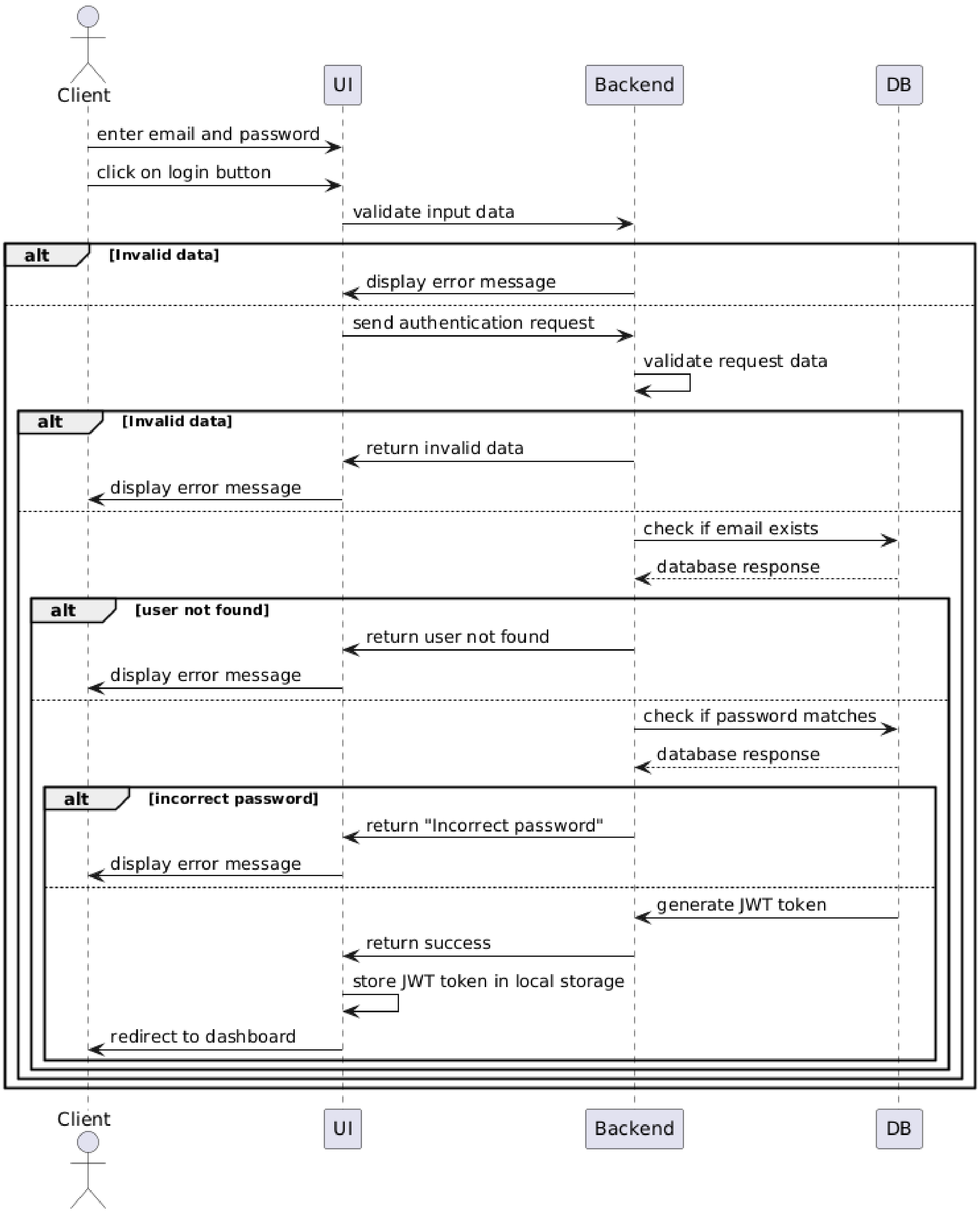

Figure 3.3: Sequence Diagram - Login

3.2.3 Implementation

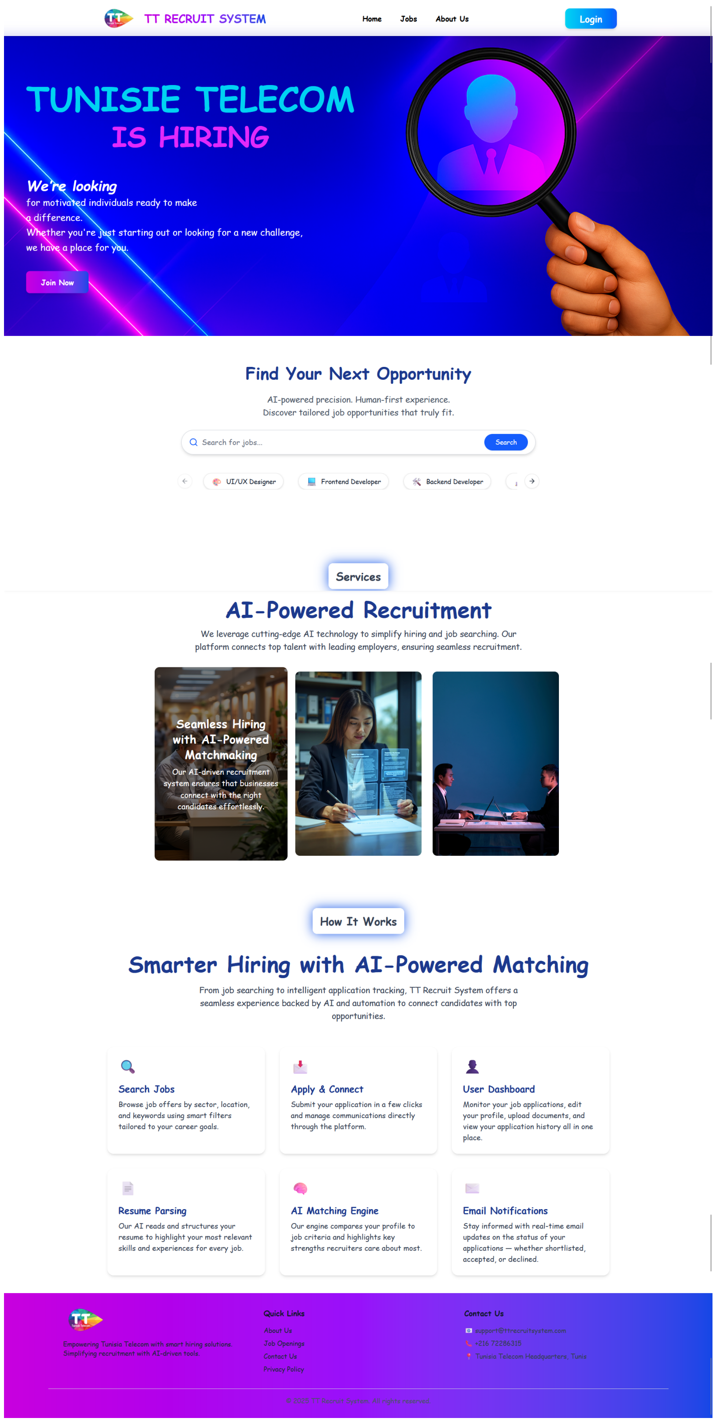

Figure 3.4: Home page

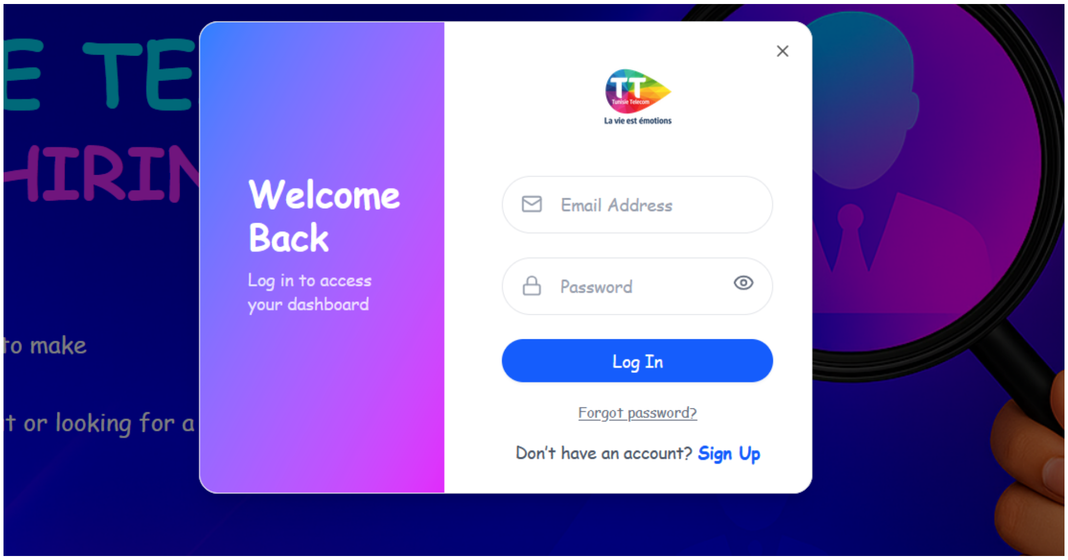

Figure 3.5: Login - interface

Figure 3.5 illustrates the Login Page of the TT Recruit System, designed for both employers and job seekers. It includes fields for email and password, a login button, and links for password recovery and account registration.

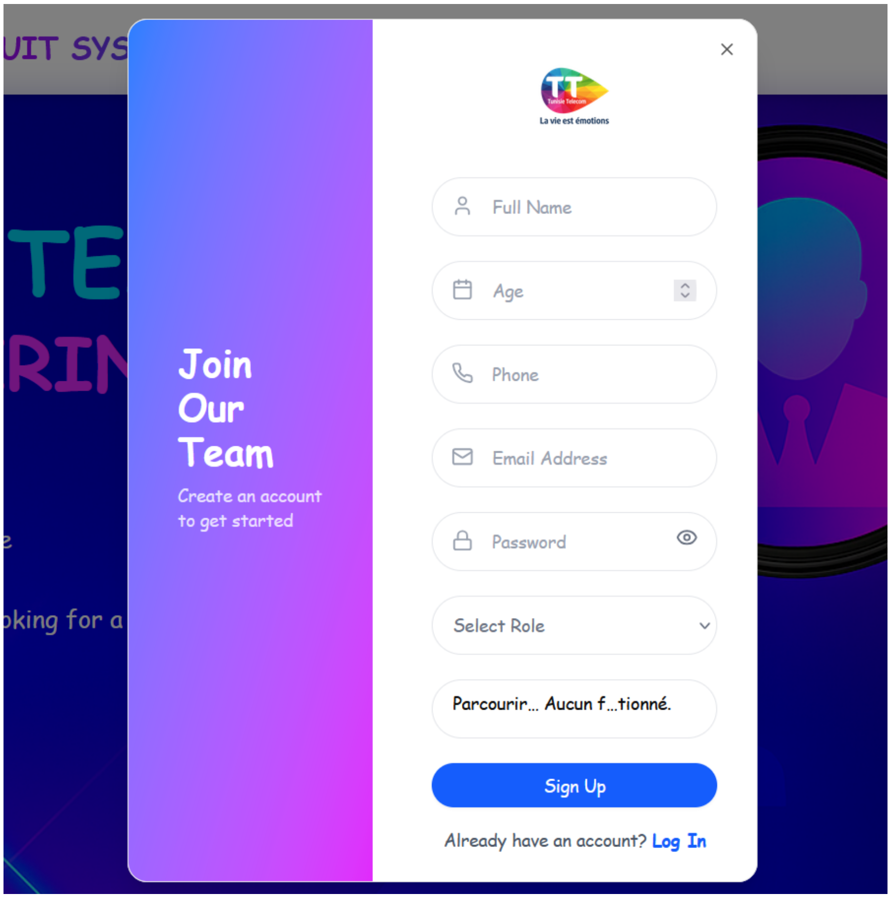

Figure 3.6: Signup - interface

Figure 3.6 shows the general signup interface for the TT Recruit System. The form includes input fields for full name, age, phone number, email, and password, along with a dropdown to select the user role (e.g., employer or job seeker). There's also an option to upload a profile image.

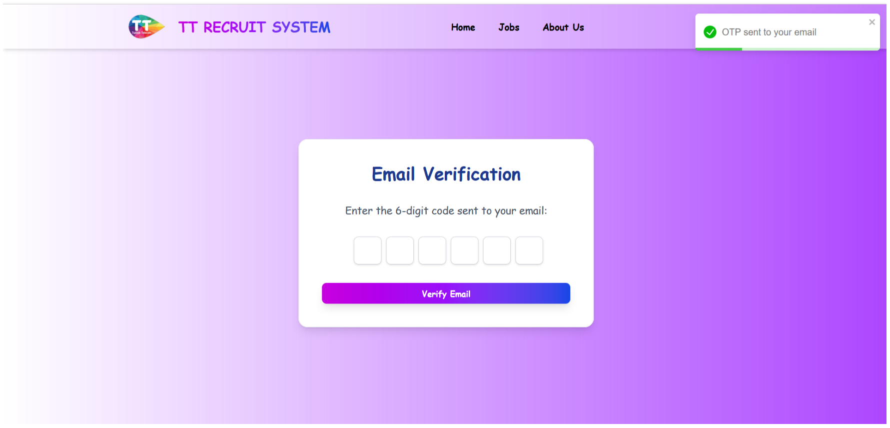

Figure 3.7: Employer email verification page

Figure 3.7 shows the Email Verification page that newly registering employers encounter after initiating the signup process on the TT Recruit System. The interface prompts the user to enter a 6-digit OTP code sent to their email to complete verification. Only after entering the correct code and clicking Verify Email will the employer's account be stored in the database. Upon successful verification the employer's account will be stored in the database and he will automatically redirected to their recruiter dashboard.

3.3 Sprint 2 : Setup The User Profile

3.3.1 Sprint Backlog

Actor	**Feature**	**User Story**
User	Profile Management	As a user, I want to view and update my profile information.
	Resume Upload	As a user, I want to upload my resume in PDF format.
	Resume Storage	As a user, I want my uploaded resume to be saved and linked to my profile.
	Profile Image Upload	As a user, I want to be able to change my profile image when i want to .

Table 3.2: Sprint 2 Backlog

3.3.2 Functional Specification

Use Case Diagram

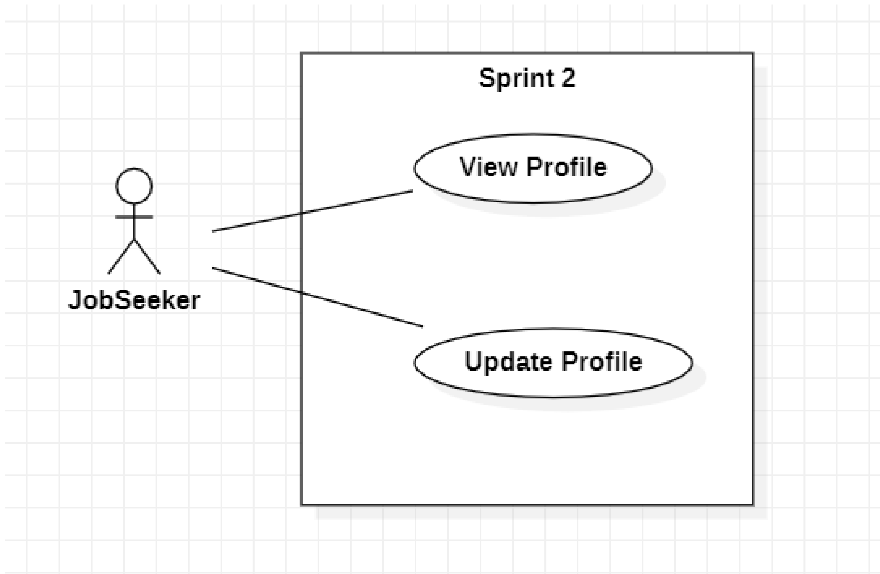

Figure 3.8: UseCase Diagram - Sprint 2

Description

- **UseCase : JobSeeker – View Profile :**

This use case allows an authenticated user to access and view their profile information. Once logged in, the user can navigate to the profile section to see personal details such as name, email, and resume information. This is a read-only action that does not involve any modification to the stored data.

- **UseCase : JobSeeker – Update Profile :**

This use case allows an authenticated user to update their profile information. The user can modify their name, change their profile image, and upload or replace their resume. The system validates the input data, stores any uploaded files, and links them to the user's profile. All changes are immediately reflected in the profile upon successful update.

Conception

- **Sequence Diagram of the Update Profile Process :**

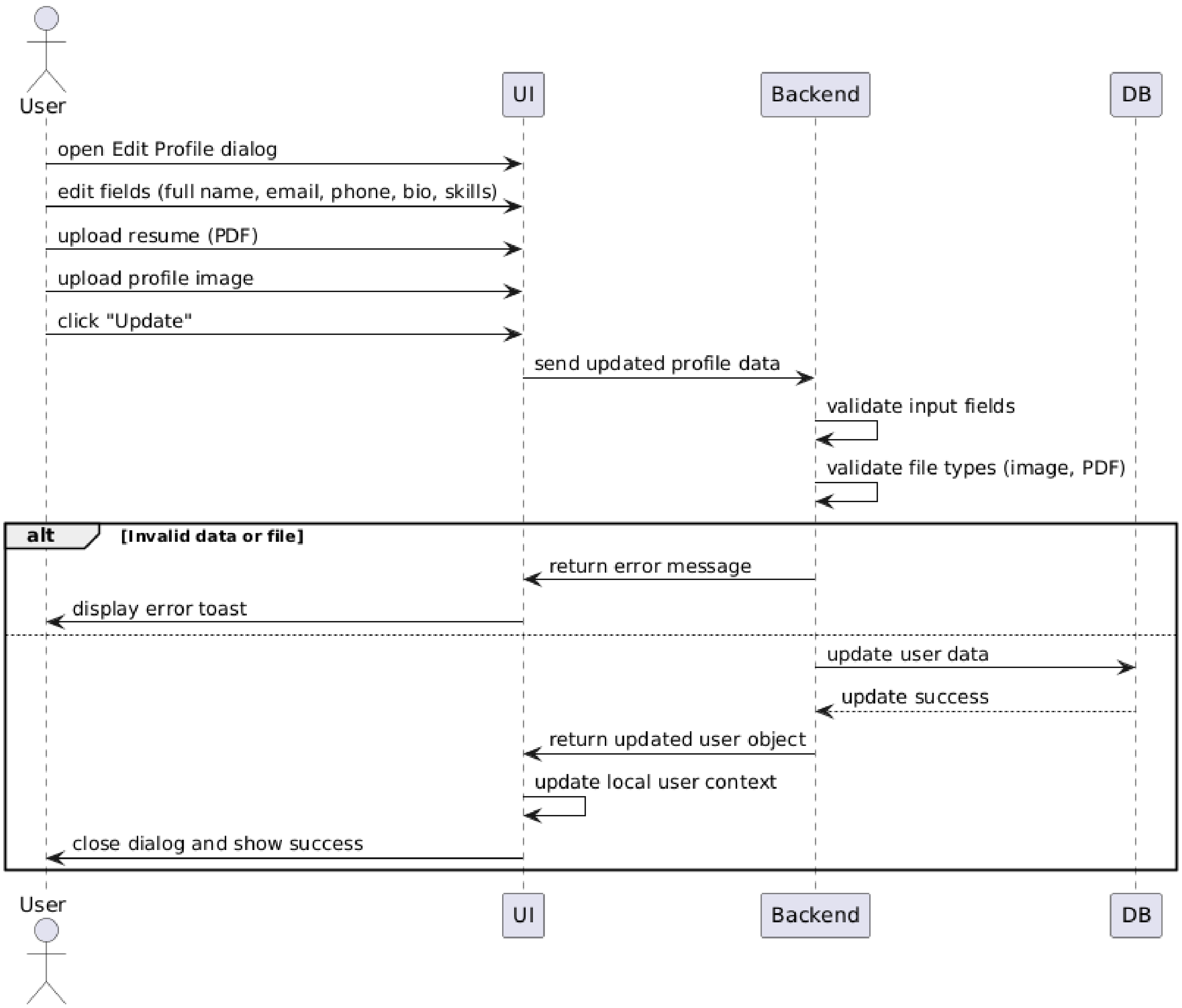

Figure 3.9: Sequence Diagram Profile Update

- **Sequence Diagram of the View Profile Process :**

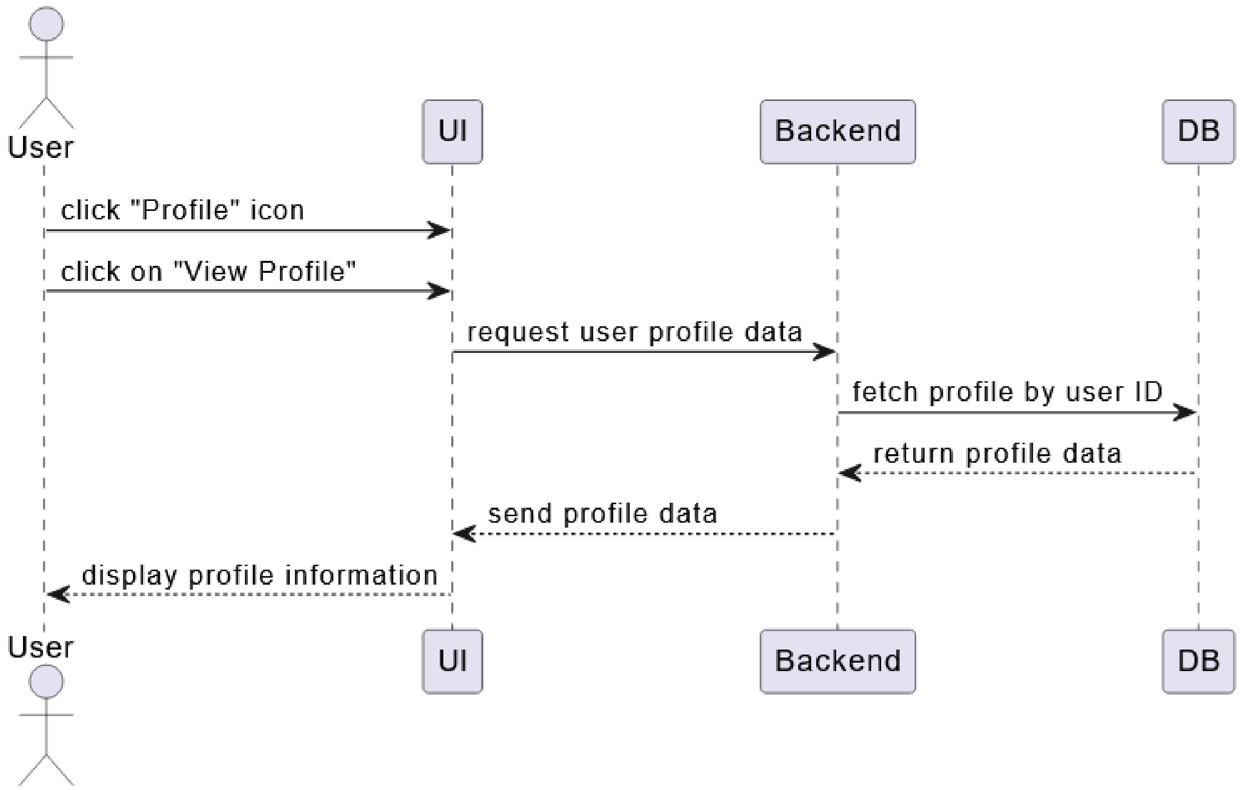

Figure 3.10: Sequence Diagram View Profile

3.3.3 Implementation

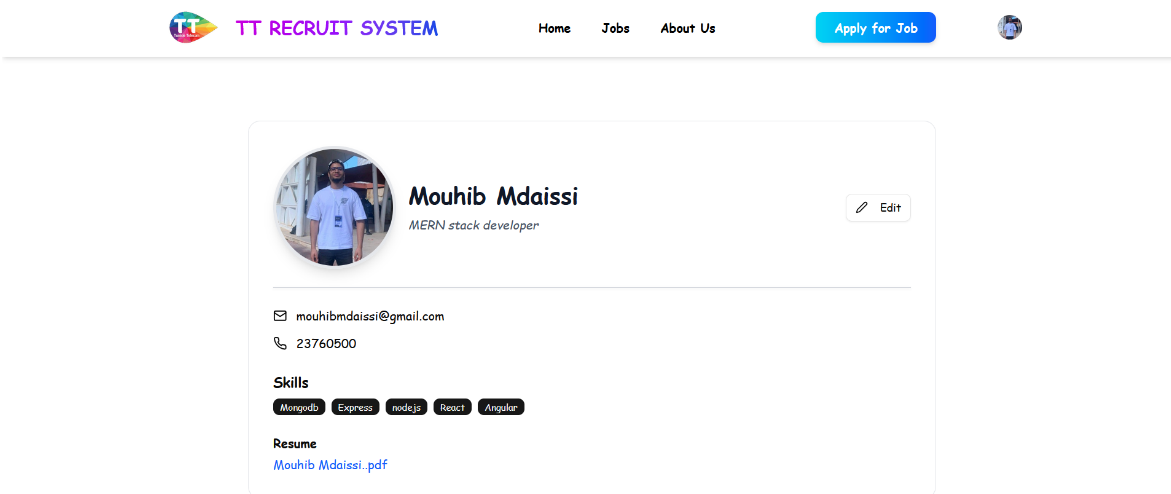

Figure 3.11: User Profile

Figure 3.11 showcases the user profile interface in the TT Recruit System. The profile card displays key information such as the user's name , bio, email, and phone number. Below that, a list of skills is shown in the form of labeled tags. The user's resume is also accessible through a clickable PDF link. An Edit button in the top-right corner of the card allows users to modify their profile details.

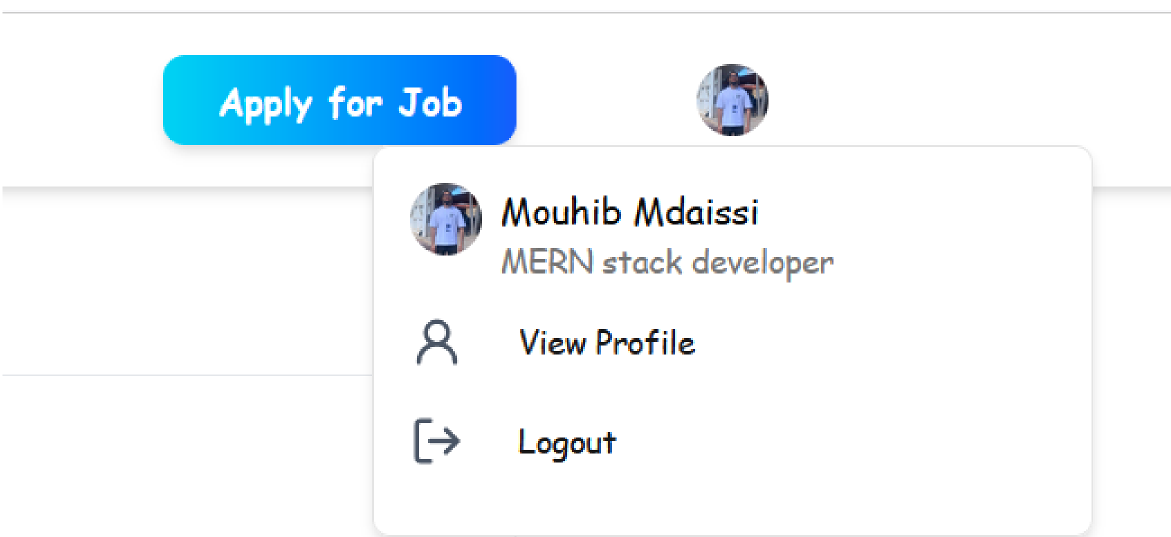

Figure 3.12: User dropdown menu showing profile overview

Figure 3.12 displays the user dropdown menu in the TT Recruit System, accessible by clicking the profile icon in the top navigation bar. The dropdown shows the user's name and bio along with two actionable options: **View Profile**, which redirects to the user's detailed profile page, and **Logout**, which signs the user out of the platform .

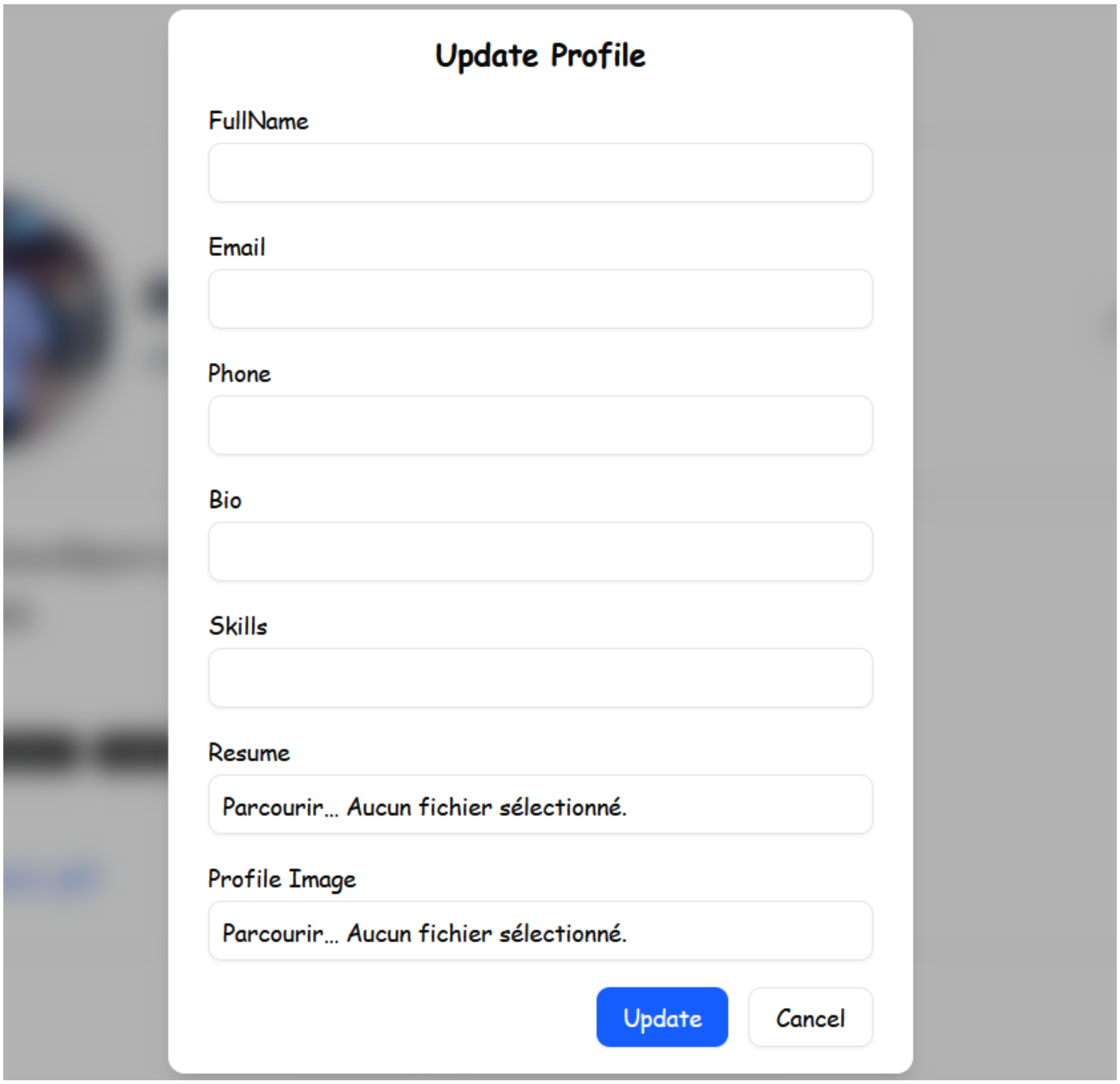

Figure 3.13: Update Profile Form

Figure 3.13 illustrates the user interface for updating a profile in the TT Recruit System. The form includes input fields for Full Name,Email, Phone, Bio, and Skills, allowing users to modify their personal and professional information. Additionally, file upload fields are provided for updating the user's Resume and Profile Image

3.4 Conclusion

In conclusion, Release 1 focused on establishing the foundational user access and profile features of the TT Recrut System. It covered the complete user identification flow including registration, role selection, and secure login, followed by the implementation of profile management capabilities such as viewing and updating personal information, uploading a resume, and adding a profile image. These core functionalities ensure that users are properly onboarded and their profiles are well structured, setting the stage for the more advanced recruiter and application features in the upcoming releases.

Chapter 4

Realease 2 :Managing Job Offers and Candidate Applications

4.1 Introduction

This chapter focuses on the implementation of the core job management and application features of the TT Recrut System. It introduces the employer-side functionality for managing job offers and the JobSeeker dashboard for browsing and applying to available positions .This release marks the beginning of direct interaction between recruiters and job seekers through structured dashboards and application tracking.

4.2 Sprint 3 : Employer Job Managment

4.2.1 Sprint backlog

Actor	Feature	User Story
Recruiter	Dashboard Access	As a recruiter, I want to access a dashboard where I can manage my job offers.
	Create Job Offer	As a recruiter, I want to create a new job offer with all necessary details.
	Manage My Job Offers	As a recruiter, I want to see and manage my job offers
	View All Job Offers	As a recruiter, I want to view all job offers posted on the platform
	View Job Description	As a recruiter, I want to view the full details of a posted job offer .
	Delete Job Offer	As a recruiter, I want to delete jobs, including expired ones, from the general job list.

Table 4.1: Sprint 3 Backlog – Employer Job Management

4.2.2 Functional Specification

Use case Diagram

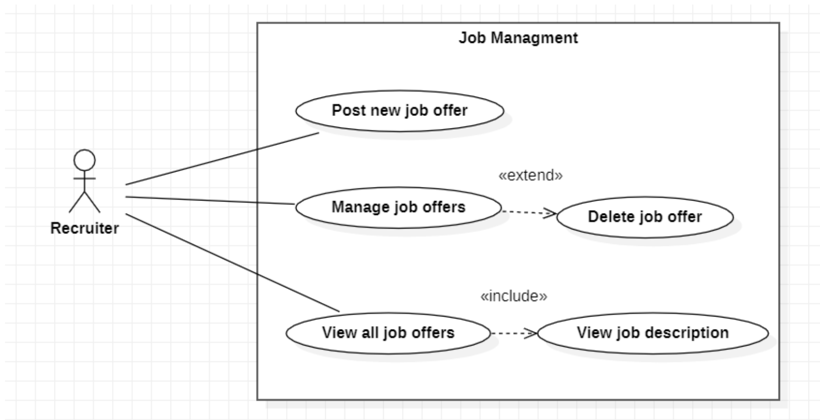

Figure 4.1: UseCase diagram - sprint3

Description

- **Use Case: Recruiter – Post Job Offer :**

This use case allows a recruiter to publish a new job offer on the platform. The recruiter must provide relevant information such as the job title, description, required skills , location , and application deadline. Once submitted, the job offer becomes visible to potential candidates and is added to the jobs list as an active job offer. The recruiter can later access and manage this posting through their dashboard

- **Use Case: Recruiter – Manage Job Offers :**

This use case allows a recruiter to access and manage the job offers they have created. The recruiter can view a list of all their posted jobs, check their status, and perform available actions on each one. Through this interface, the recruiter maintains control over the visibility and accuracy of the listings. The manage job offers interface also provides access to delete actions if needed.

- **Use Case: Recruiter – Delete Job Offer :**

This use case allows a recruiter to delete a job offer they have previously created. It is triggered from within the Manage Job Offers interface, where the recruiter can view and manage job postings. This option is typically used when a job is filled or no longer open for applications.

- **Use Case: Recruiter – View all Job Offers :**

This use case allows a recruiter to view all job offers published by the company on the platform. The recruiter can navigate to a dedicated interface that displays every active posting, regardless of who created it. This read only view helps recruiters stay informed about all current openings and maintain awareness of ongoing recruitment needs within the organization.

- **Use Case: Recruiter – View Job Description :**

This use case allows a recruiter to view the full details of a job offer. After selecting a specific job from the list, the system retrieves the job's description, including its title, mission, skills and qualification, location, contract type,posted and closing dates.It is a read-only operation and does not allow any modifications.

Conception

- **Sequence diagram of the post job process :**

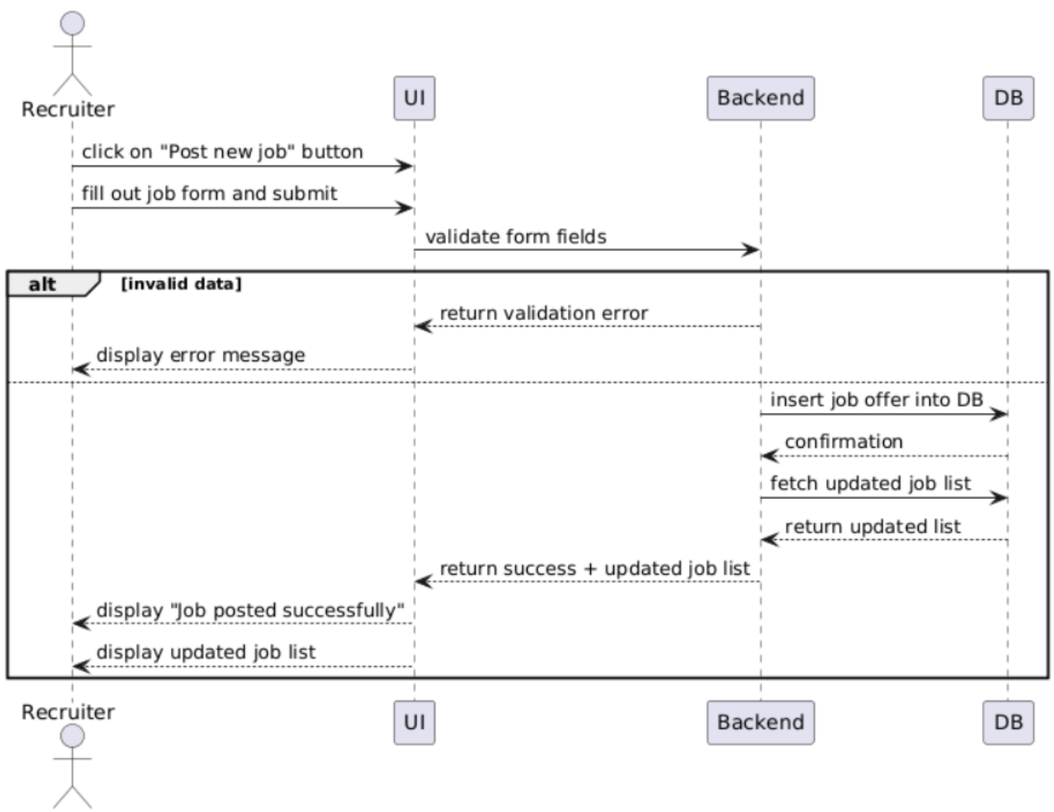

Figure 4.2: Sequence diagram - Post job

- **Sequence diagram of the manage job process :**

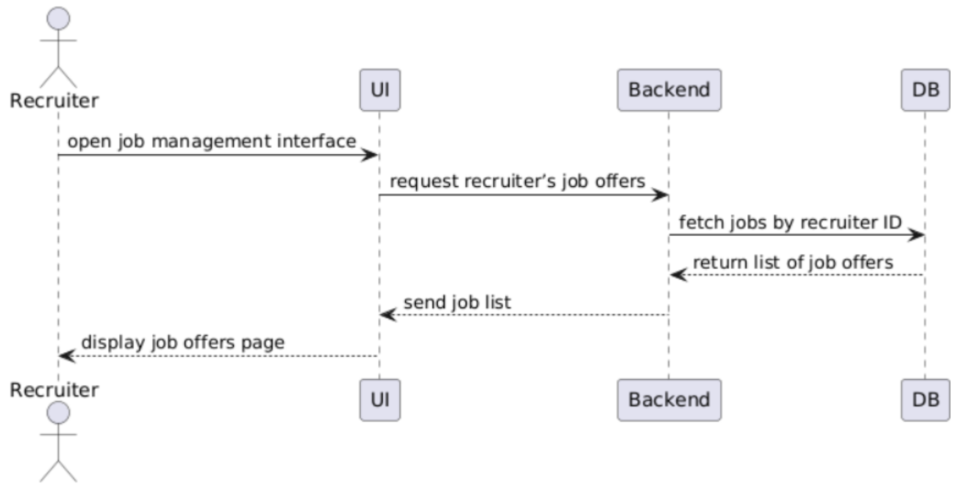

Figure 4.3: Sequence diagram - Manage jobs

- **Sequence diagram of the delete job process :**

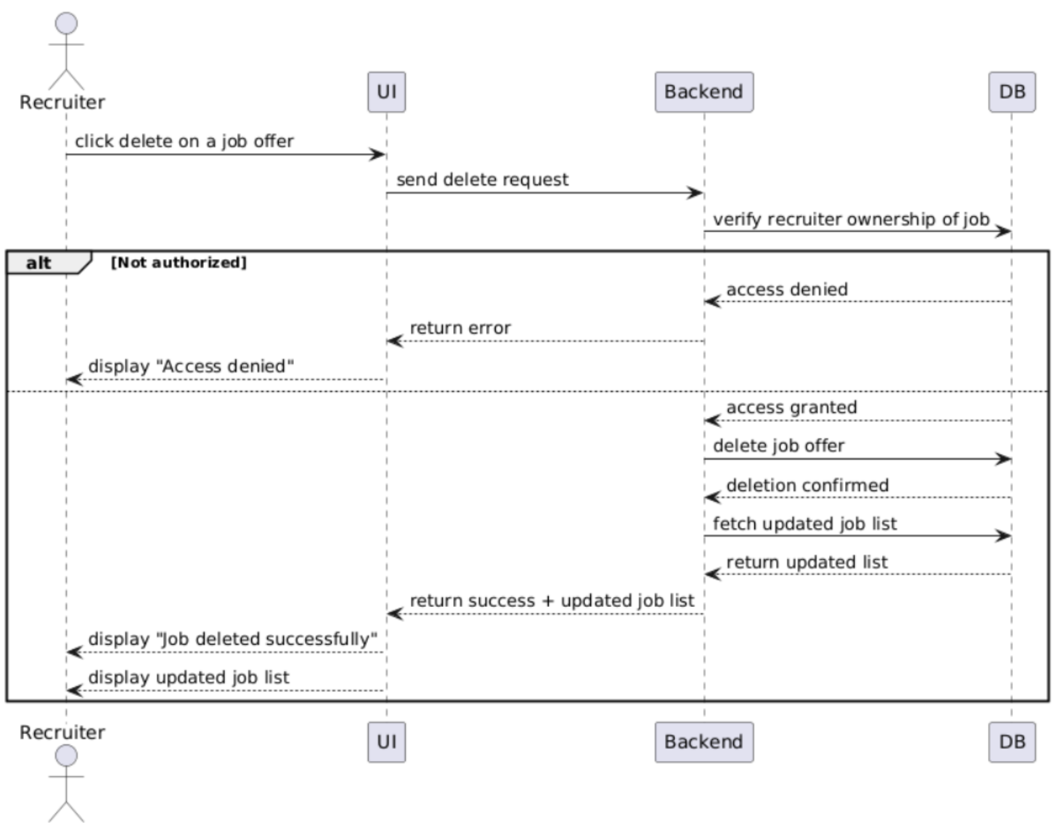

Figure 4.4: Sequence diagram - Delete job

- **Sequence diagram of the view all job offers process :**

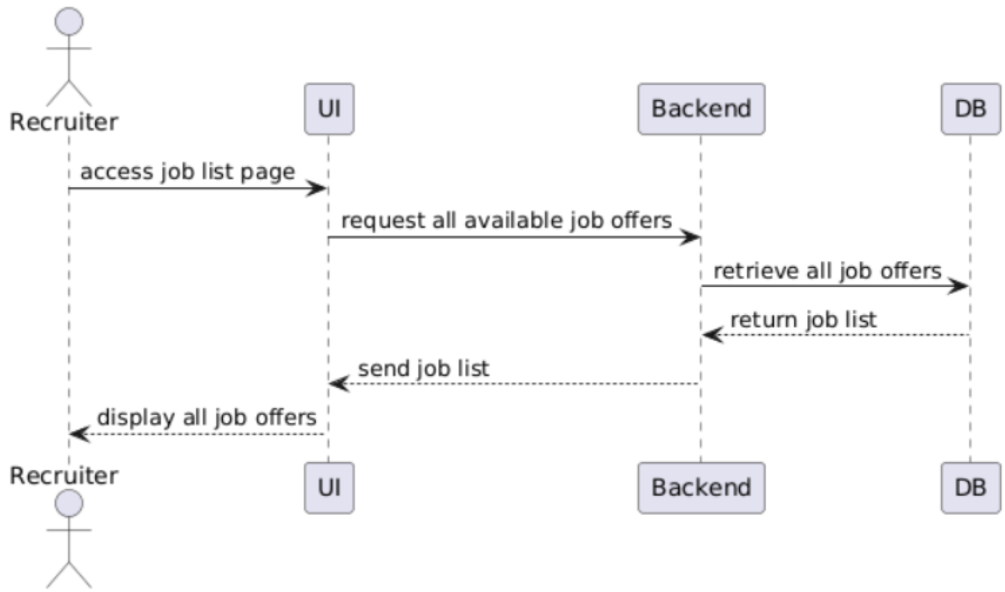

Figure 4.5: Sequence diagram - View job offers

- **Sequence diagram of the view job description process :**

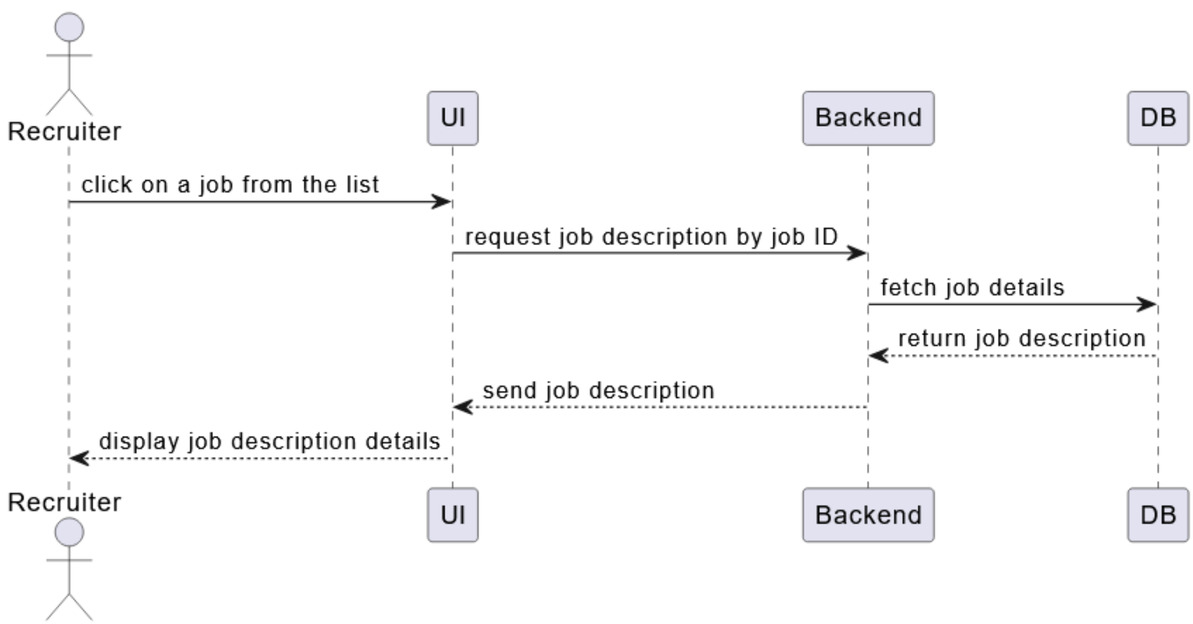

Figure 4.6: Sequence diagram - View job description

4.2.3 Implementation

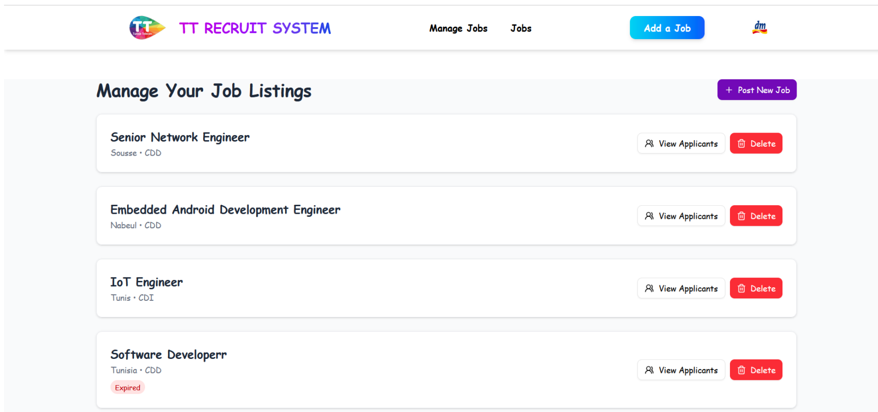

Figure 4.7: Manage jobs interface

Figure 4.7 displays the **Manage Job Listings** interface from the recruiter dashboard in the TT Recruit System. The section provides a list of all posted job offers even expired ones by the logged in recruiter .Recruiters can perform two key actions per listing: **View Applicants** to see who applied, and **Delete** to remove the listing .A **Post New Job** button at the top right allows recruiters to add new openings, streamlining job management in a clear, actionable layout.

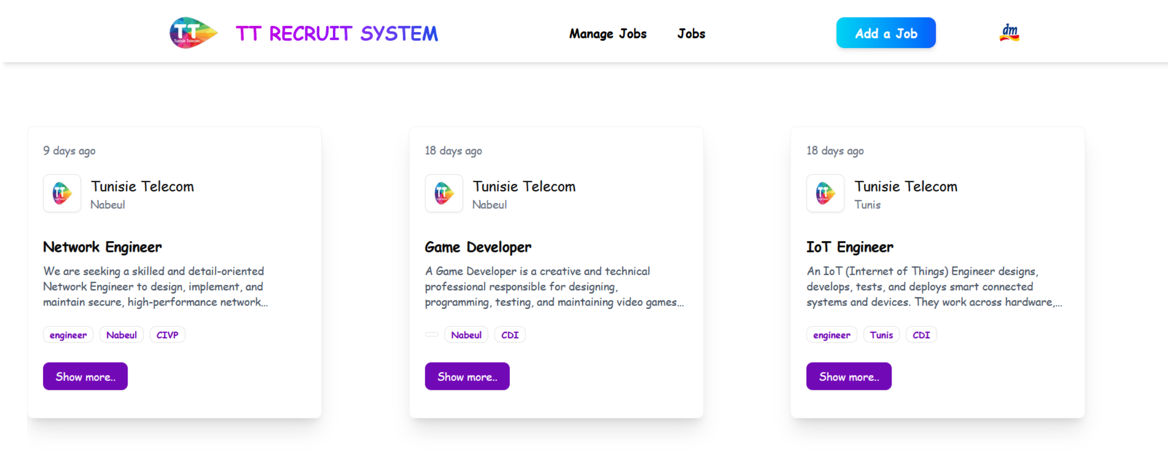

Figure 4.8: View Job Offers

Figure 4.8 presents the interface for viewing available job offers within the recruiter dashboard of the TT Recruit System. The display features job cards Each card includes a "Show more" button for viewing full details.

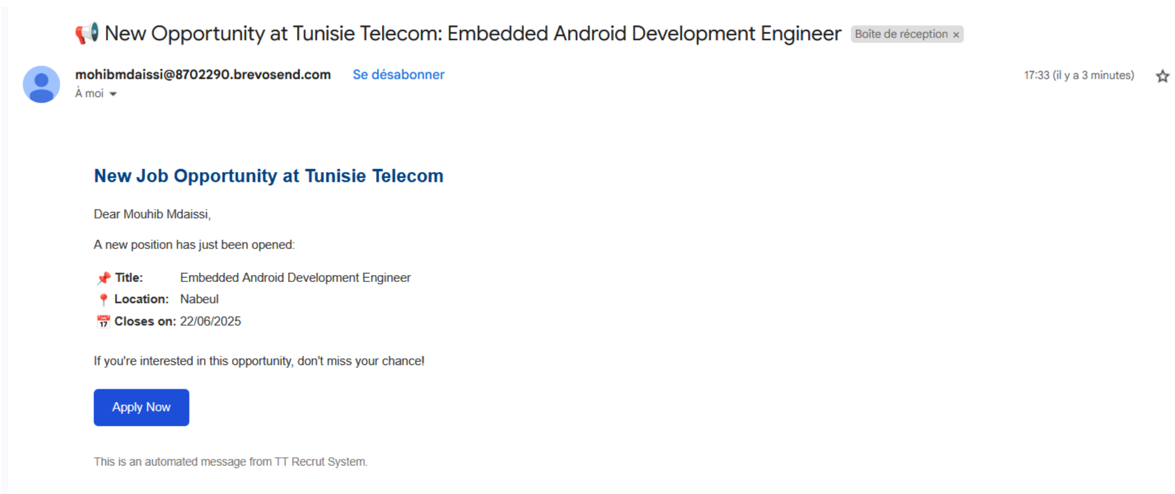

Figure 4.9: Interface of emaill notification sent when a new job is posted

figure 4.9 showcases the email notification received by a user when a new job is posted on the TT Recruit System. The email serves as an alert to keep users informed of newly available positions, encouraging immediate engagement with relevant opportunities.

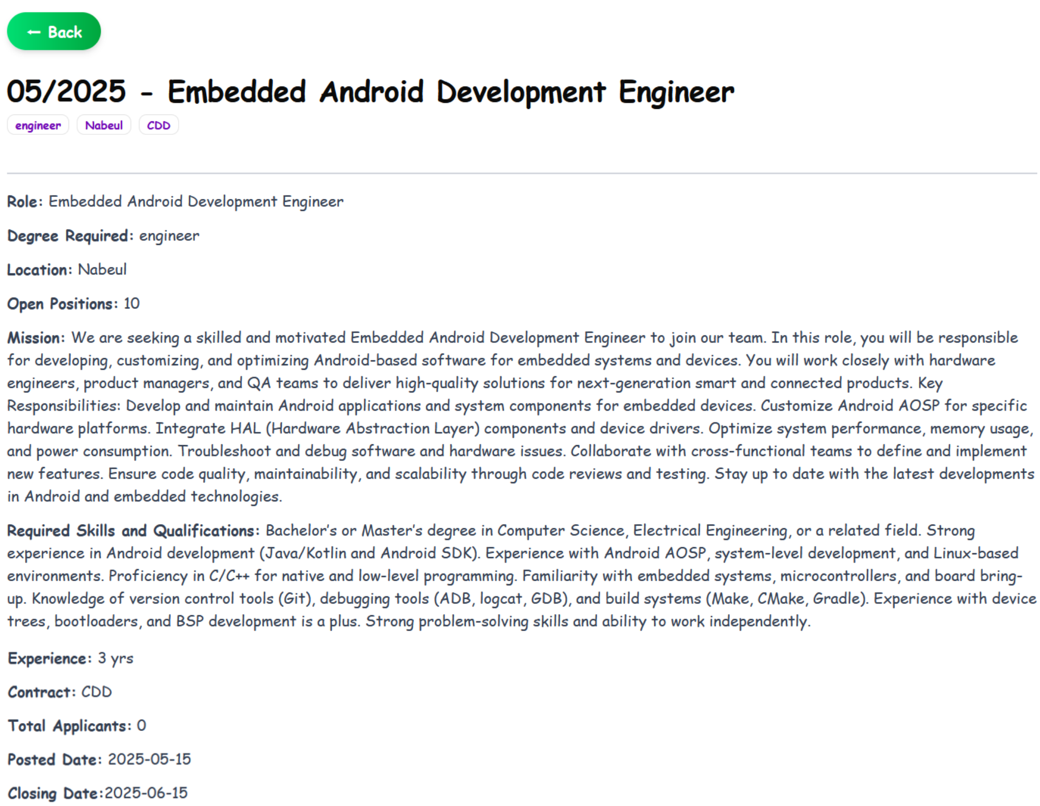

Figure 4.10: View job details

Figure 4.10 displays the detailed view of a job posting within the TT Recruit System. The page provides comprehensive information including the role, required degree, location and number of open positions. A detailed **mission** and **responsibilities** section outlines the expected tasks.The **requirements** section lists preferred qualifications and skills. Additional metadata such as the number of applicants, contract type, posting date, and closing date are also included.

4.3 Sprint 4 : Job Application Management and JobSeeker Interface

4.3.1 Sprint Backlog:

Actor	**Feature**	**User Story**
JobSeeker	JobSeeker Dashboard	As a job seeker, I want to see the latest job offers and search/filter available jobs.
	Search and Filter jobs	As a job seeker, I want to search and filter jobs by their location and title
	Apply to Job	As a job seeker, I want to be able to apply to active jobs
	Resume Selection	As a job seeker, I want to choose whether to upload a new resume or use the one in my profile.
	Application Tracking	As a job seeker, I want to view the status of my appication
Recruiter	View Applicants per Job	As a recruiter, I want to view the list of candidates who applied for each of my job offers.
	Update Application Status	As a recruiter, I want to update the application status (accepted, rejected, or shortlisted).

Table 4.2: Sprint 4 Backlog – Job Application Management and JobSeeker Interface

4.3.2 Functional Specification

Use Case Diagram

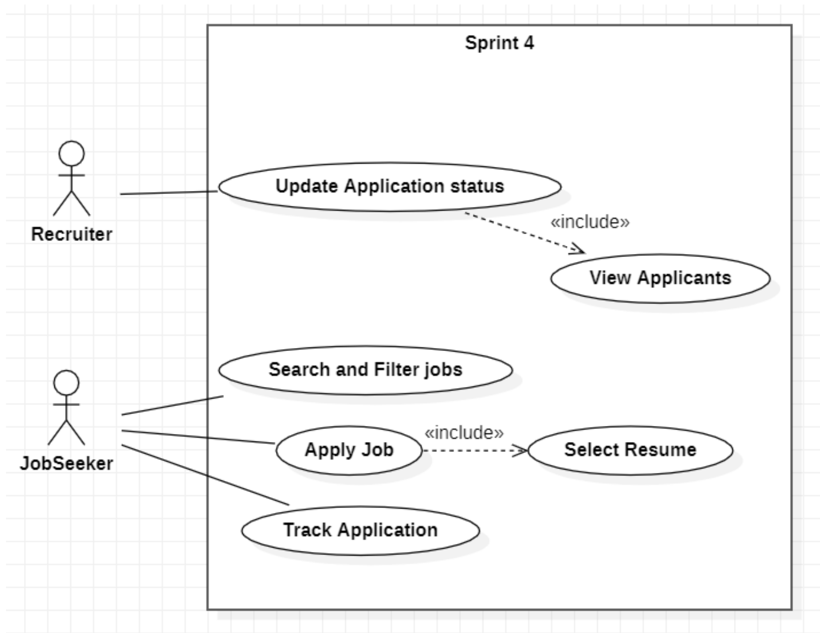

Figure 4.11: Sprint4 - useCaseDiagram

Description

- **Use Case: Jobseeker – Apply to Job :**

This use case allows a job seeker to submit an application for a job offer directly from the job listing page. By clicking the "Apply" button, the system checks eligibility and processes the application request. Before proceeding, the system ensures that the job seeker has no active or pending application. A job seeker can only apply to one job at a time, and is allowed to submit a new application only if the previous one has been rejected. This ensures that each candidate is evaluated for a single position at a time. This use case includes the resume selection step as a required sub action.

- **Use Case: JobSeeker – Select Resume :**

This use case allows a job seeker to choose how they want to attach a resume during the job application process. The job seeker can either upload a new resume or reuse the one stored in their profile. This step is mandatory and is included in the “Apply to Job” use case.

- **Use Case: JobSeeker – Track Application:**

This use case allows a job seeker to track the progress of their job application. After applying to a job, the user can view the current status (e.g., pending, reviewed, accepted, rejected) through their dashboard. This is a read-only action that retrieves application data.

- **Use Case: Recruiter – View Applicants:**

This use case is triggered when a recruiter clicks the “View Applicants” button associated with a specific job posting. Upon activation, the system loads the list of job seekers who have applied for that particular job and redirects the recruiter to a dedicated Applicants page. This page displays application details such as candidate names, resumes, and current status. This use case is a prerequisite for performing actions like updating application statuses.

- **Use Case: Recruiter – Update Application Status :**

This use case allows a recruiter to change the status of a candidate’s application. The recruiter can mark an application as shortlisted, accepted, or rejected. The application status begins as **Pending** by default. The recruiter can change the status to **Shortlisted**. From **Shortlisted**, the recruiter must choose a final decision: either **Accepted** or **Rejected**. Once an application reaches **Accepted** or **Rejected** status, it becomes final and cannot be modified further. This action extends the “View Applicants” use case, as the recruiter must first access the application list before making any changes.

- **Use Case: JobSeeker – Search and Filter jobs:**

A job seeker can effectively search and filter job postings using this use case by applying particular criteria. The job seeker can use the dashboard’s search field to enter a job title or a portion of it, and the system will look for job titles that match. The system will reveal the applicable job(s) details if a match is detected; if not, it will display a "no jobs found" message. Additionally, a filter card area on the jobs page allows the job seeker to choose a **location**, **industry**, or both to filter job postings. The system shows jobs that fit the chosen location, industry, or both by enabling filtering by both criteria. Because of the increased flexibility this offers, job seekers can find what they are looking for .

Conception

- **Sequence diagrams of the Search & Filter process :**

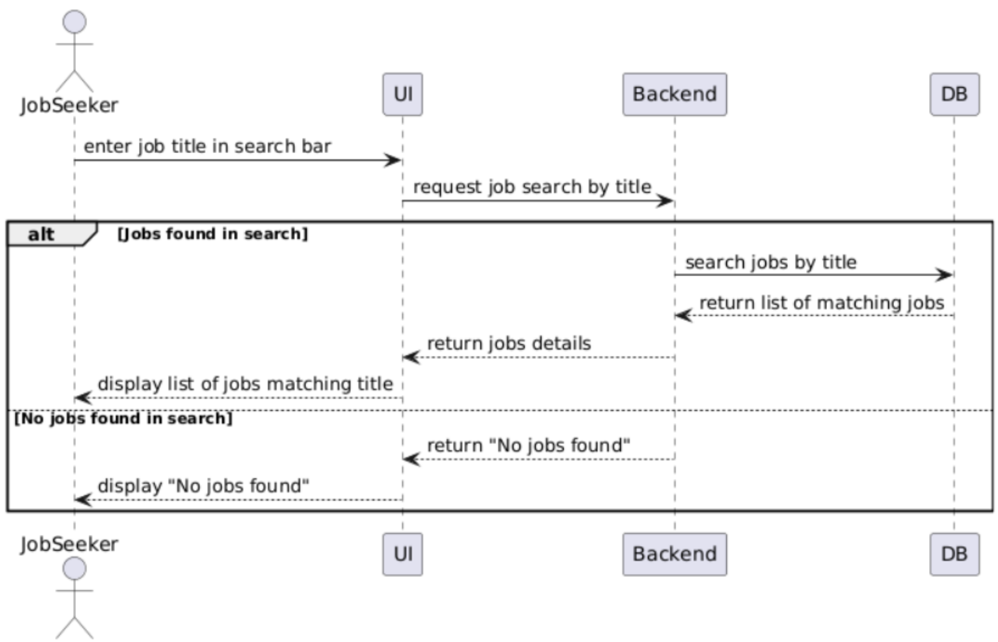

Figure 4.12: Sequence Diagram - Search job

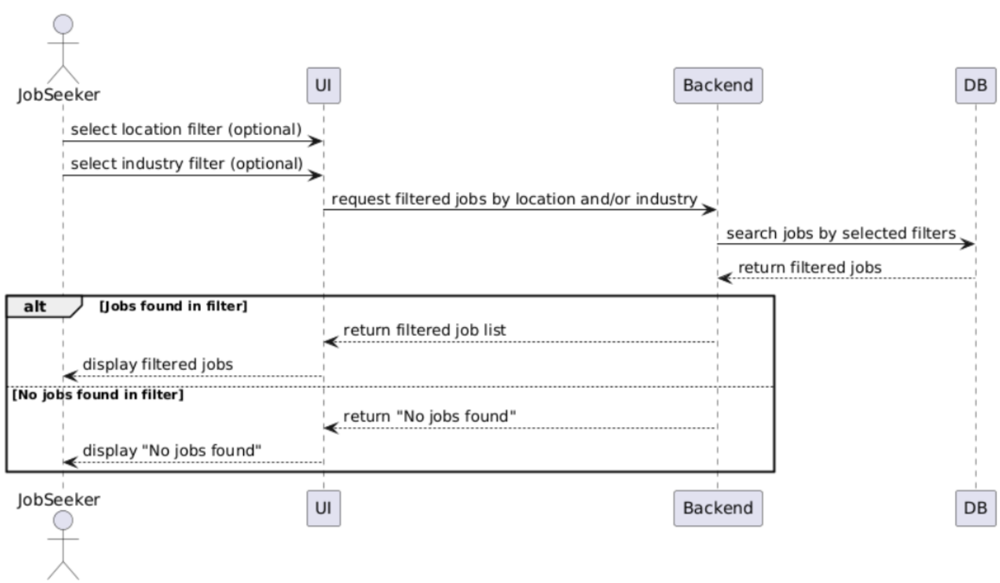

Figure 4.13: Sequence Diagram - Filter jobs

- **Sequence diagram of the apply to job process :**

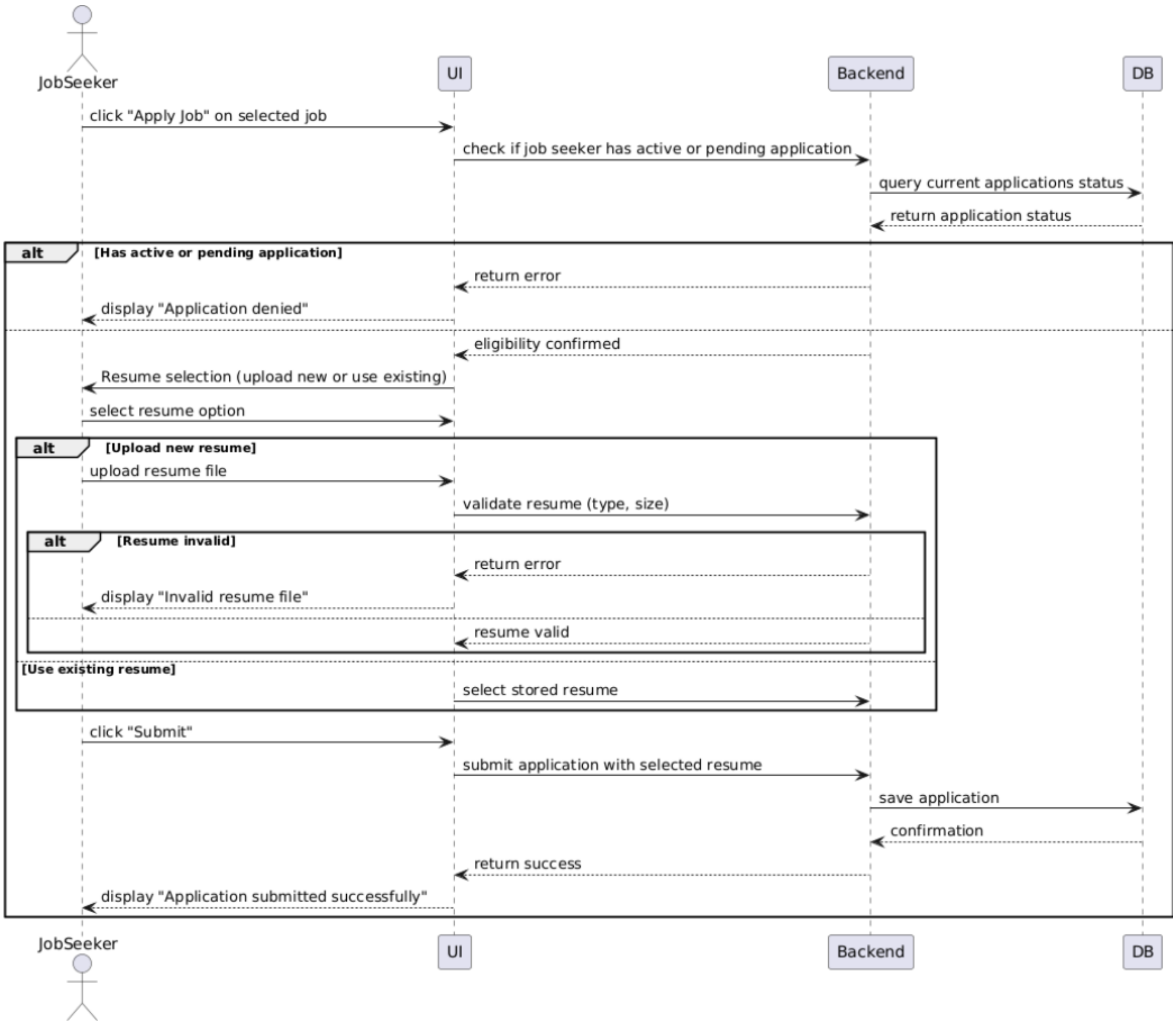

Figure 4.14: Sequence diagram - Apply to job

- **Sequence diagram of the track application process:**

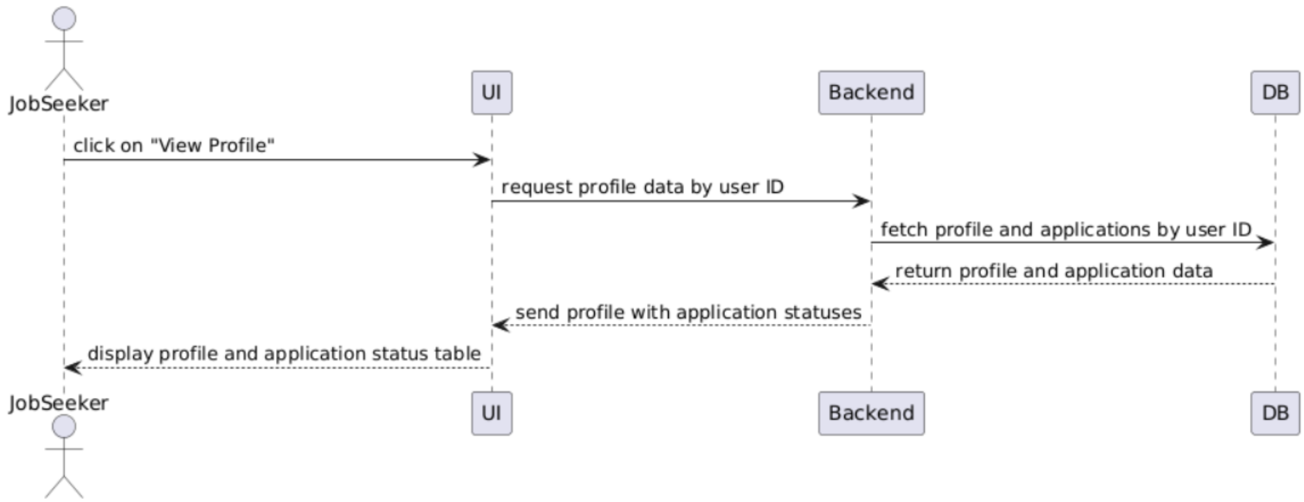

Figure 4.15: Sequence diagram - Track application

- **Sequence diagram of the view applicants process :**

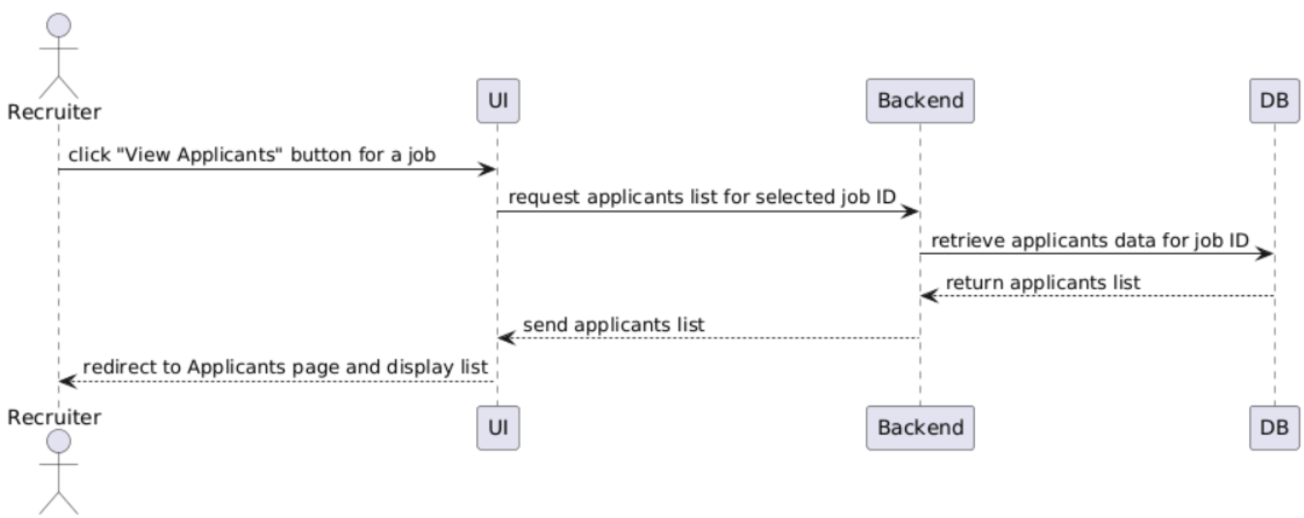

Figure 4.16: Sequence diagram - View applicants

- **Sequence diagram of the update application status process :**

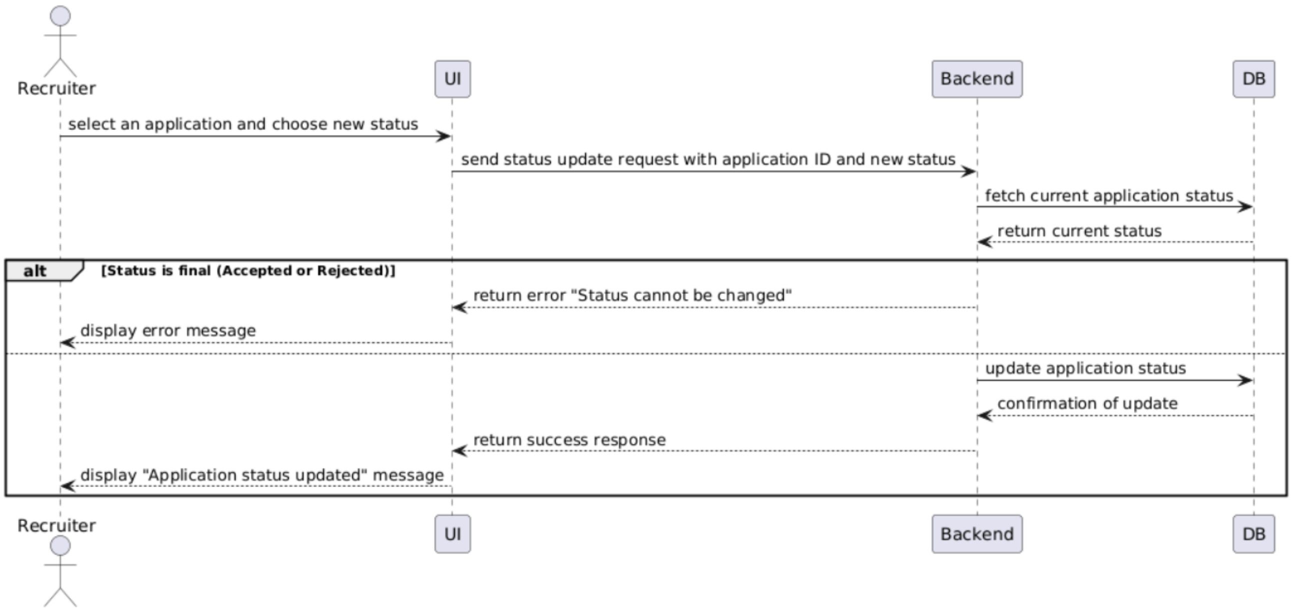

Figure 4.17: Sequence diagram - Update application status

4.3.3 Implementation

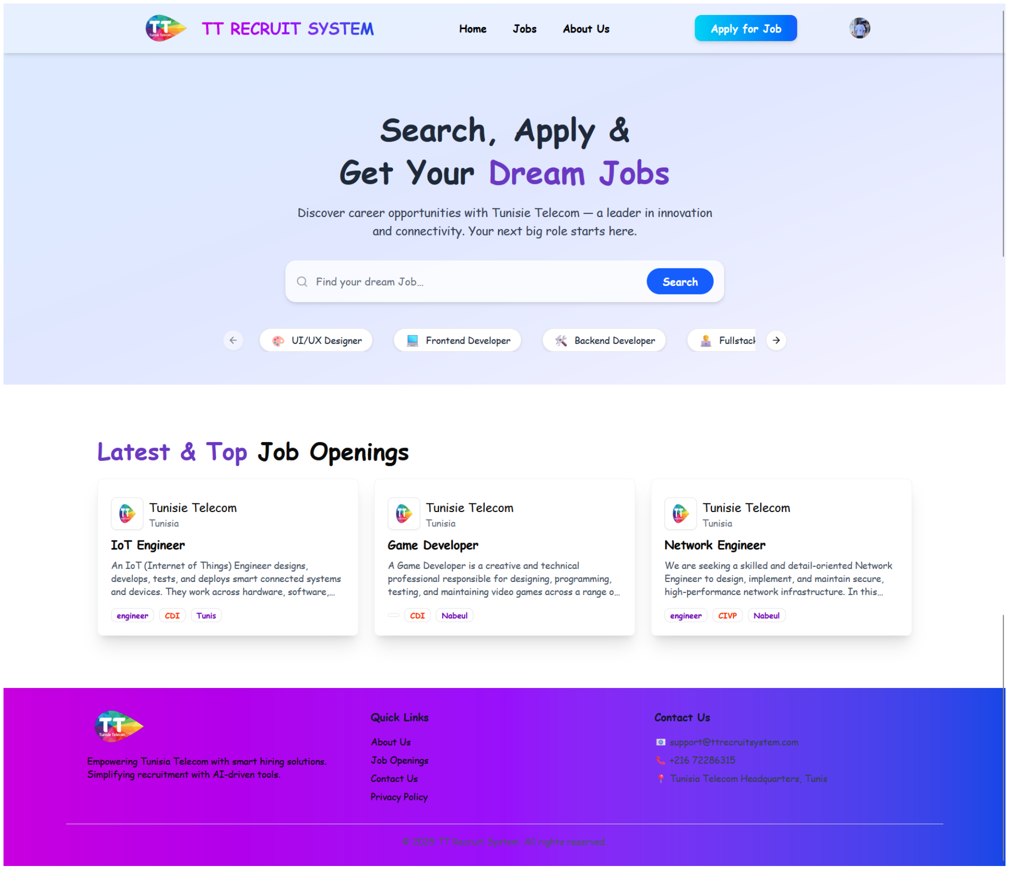

Figure 4.18: jobSeeker Dashboard

Figure 4.18 presents the main dashboard interface for job seekers within the TT Recruit System. The top section features a prominent search bar encouraging users to explore opportunities by job title . Below the search area, a row of filter buttons allows quick category-based exploration. The middle section displays a curated list of the Latest & Top Job Openings, showcasing multiple job cards with key details. The footer provides quick links and contact information, completing a clean and informative homepage for job seekers.

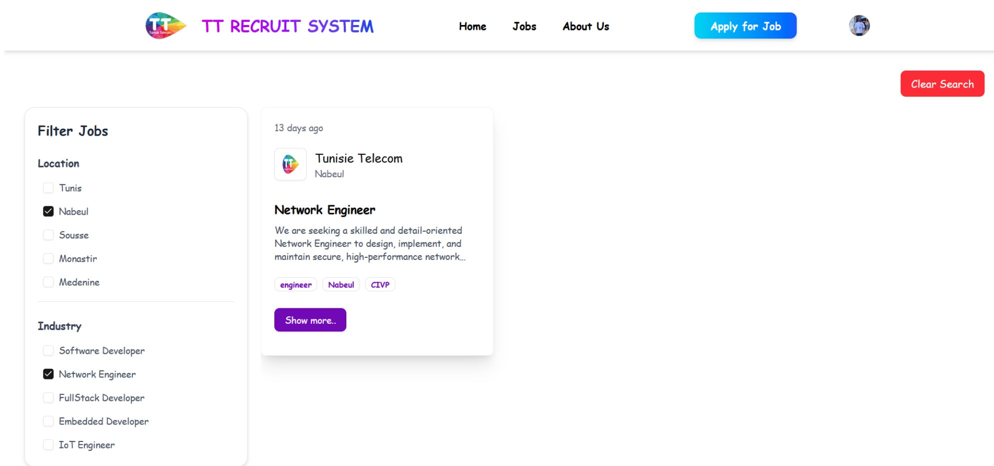

Figure 4.19: Job results filtered using the filter panel

Figure 4.19 displays the job search interface within the TT Recruit System, showcasing how job seekers can refine job listings using a filter panel. The panel on the left allows filtering based on **location** and **industry**, This interactive filtering mechanism enables users to quickly narrow down job offers that match their preferences and qualifications.

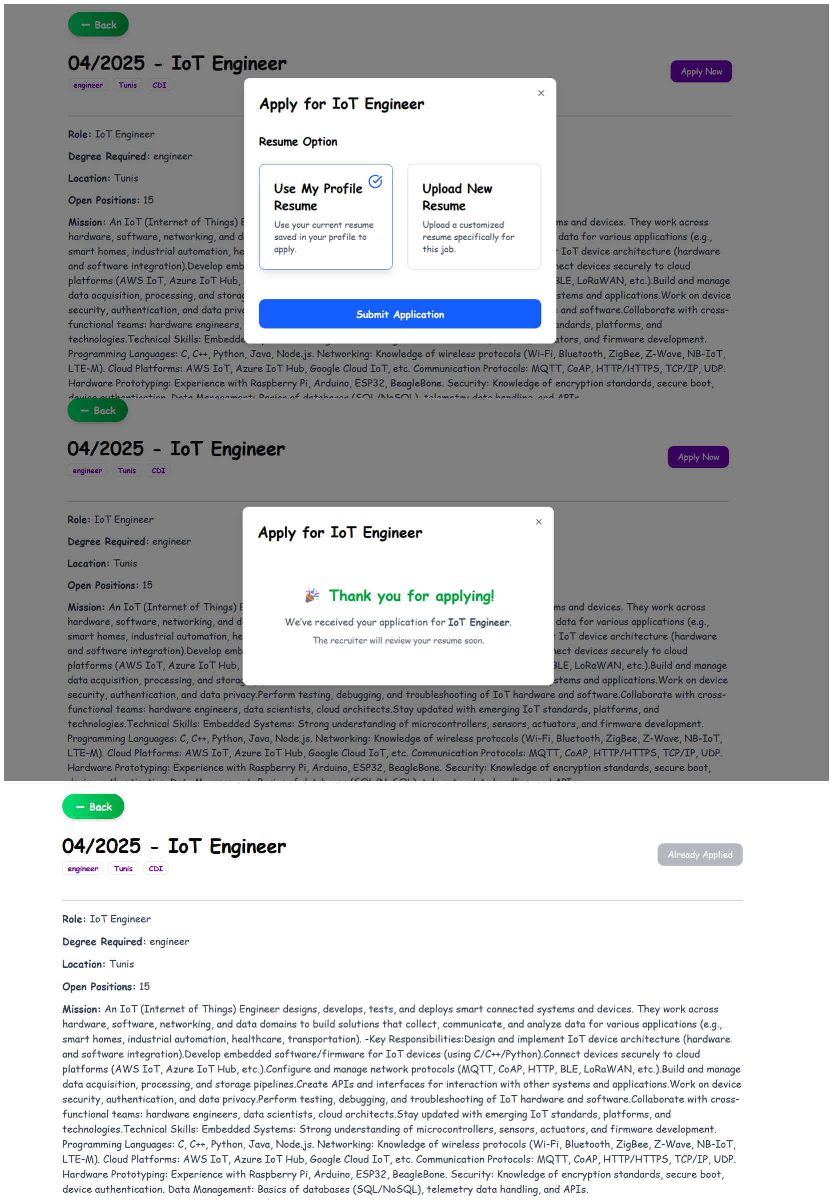

Figure 4.20: Job application process – successful submission flow.

Figure 4.20 illustrates the complete and successful job application process for a job in the TT Recruit System. The first step shows a modal where the user selects a resume submission method either using the existing profile resume or uploading a new one , before clicking the **Submit Application** button. The next view confirms the successful submission with a message: **"Thank you for applying!"**, reassuring the user that their application has been received. Finally, the job listing interface updates to reflect the applied state, disabling the "Apply Now" button and preventing multiple submissions. This flow ensures a clear and user-friendly application experience.

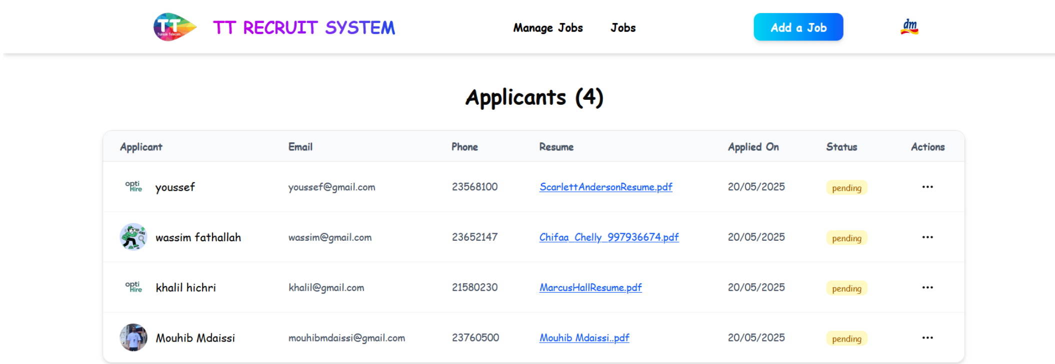

Figure 4.21: View Applicants interface

Figure 4.21 shows the "View Applicants" interface of the TT Recruit System, where recruiters can see a list of all candidates who applied for a job. The table displays key information for each applicant, including their name, email, phone number, resume link, application date, and current application status .

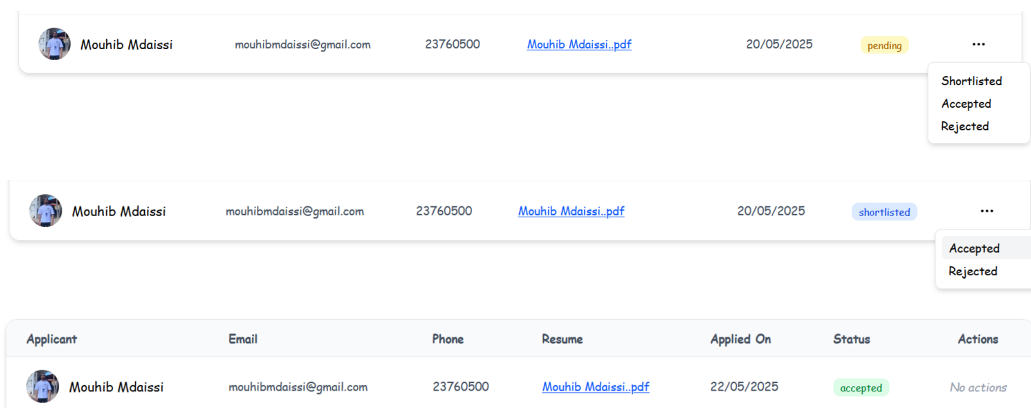

Figure 4.22: Status Updated

Figure 4.22 presents the status progression of a job application within the recruitment system. It showcases three distinct stages: initially, the application appears with a **pending** status, allowing the recruiter to take action by selecting one of the available options—**Shortlisted**, **Accepted**, or **Rejected**. Once the recruiter selects **Shortlisted**, the status updates accordingly, reflecting the candidate's progression in the hiring process. Finally, upon acceptance, the status changes to **Accepted**, highlighted in green, and the action buttons are disabled, indicating that no further action is required. This interface provides a clear and interactive way to track and manage application decision.

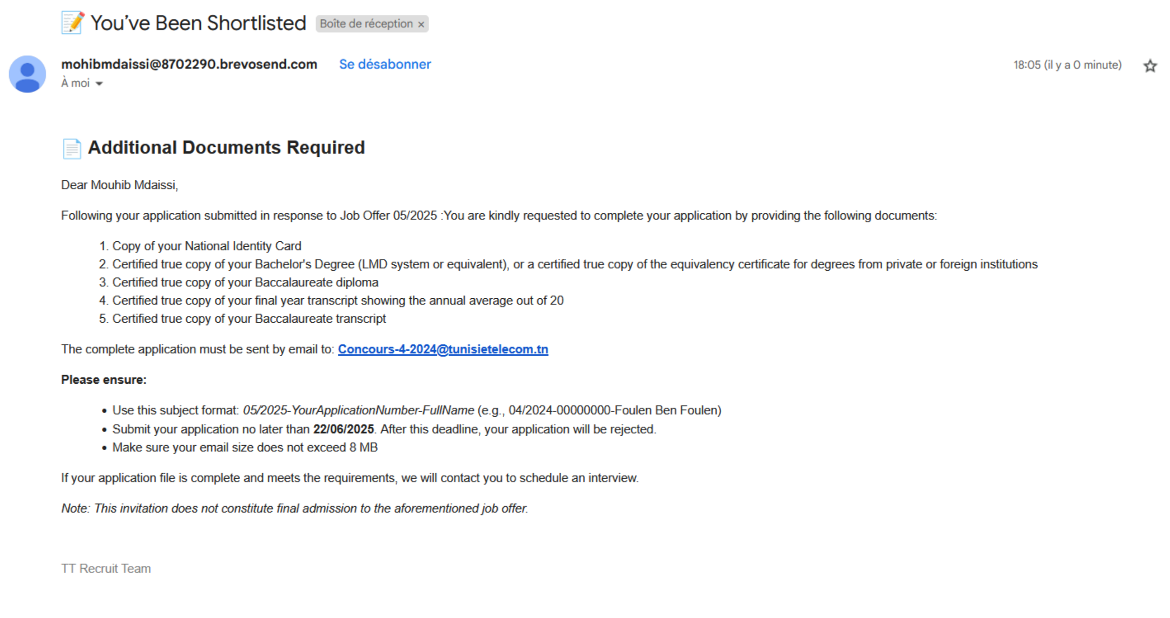

Figure 4.23: Interface of the email notification received when the application status is updated

Figure 4.23 shows the interface of an email notifying the applicant , that they have been shortlisted for a job position. The email requests the submission of additional documents to proceed with the application .

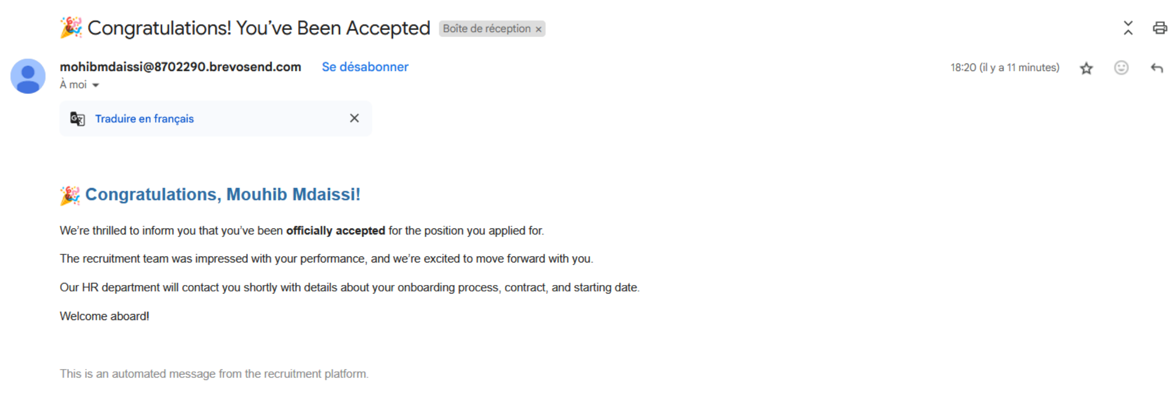

Figure 4.24: Acceptance email

Figure 4.24 shows an automated acceptance email sent to the applicant . Confirming the candidate's official acceptance for the position applied for .

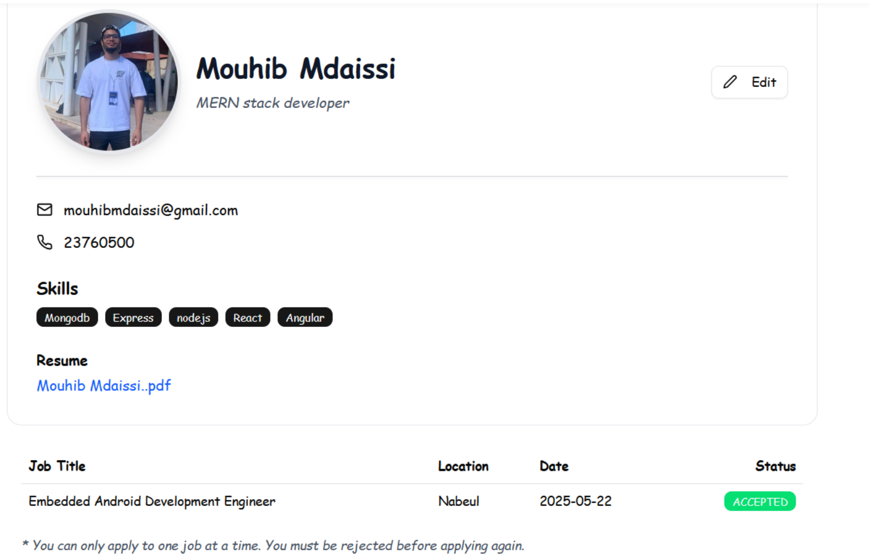

Figure 4.25: Track Application table

Figure 4.25 presents a job application tracking table showing key details of the user's current application. It includes the job title , location, and application date . The status column that indicates the application status (accepted , rejected, shortlisted)

4.4 Conclusion

In conclusion , Release 2 established the core recruitment flow, enabling recruiters to manage job offers and job seekers to apply, track their application, and interact with job listings. It laid the foundation for meaningful interaction between both user roles.

Chapter 5

Release 3 : AI-Based Resume Matching System

5.1 Introduction

In the context of an intelligent recruitment system, it is essential to move beyond basic keyword matching and implement a method capable of understanding the semantic meaning of text. The AI module in this system is designed to rank candidates based on how closely their resumes match job descriptions, using both skill-based comparison and semantic similarity.The AI module is based on a pretrained **sentence-level BERT model**
(all-MiniLM-L6-v2) which allows the system to evaluate semantic similarity between job descriptions and candidate resumes.

5.2 Technical Terminology

5.2.1 BERT (Bidirectional Encoder Representations from Transformers)

BERT is a deep learning language model developed by Google that understands text by processing it bidirectionally. It is pre-trained on large datasets and fine-tuned for tasks like classification, question answering, and text similarity. [20]

5.2.2 Embedding

An embedding is a numeric vector that represents the semantic content of text. Word and sentence embeddings allow machine learning models to compare text meaning in a mathematically meaningful way. [21]

5.2.3 Semantic Similarity

Semantic Similarity is the task of determining how similar two sentences are, in terms of what they mean . In this system, BERT-based sentence embeddings are used to compute similarity between resumes and job descriptions. [22]

5.2.4 Cosine Similarity

Cosine similarity is a metric that quantifies how similar two vectors are by measuring the cosine of the angle between them. It evaluates whether the vectors point in the same direction, regardless of their magnitude. In text analysis, it is commonly used to assess the similarity between documents or sentence embeddings based on their semantic content. [23]

$$similarity(A, B) = cos(\theta) = \frac{A \cdot B}{||A||\ ||B||},$$

Figure 5.1: Cosine Similarity Formula

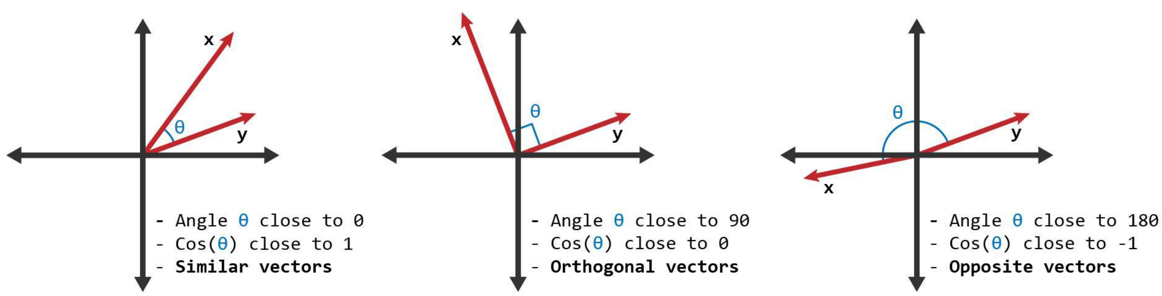

Figure 5.2: Geometric Interpretation of Cosine Similarity Between Vectors
[24]

Figure 5.2 illustrates the geometric interpretation of cosine similarity : Vectors with a small angle between them have high similarity , orthogonal vectors have no similarity and opposite vectors have negative similarity

5.3 Words Vectorization Techniques:

To enable automated comparison between resumes and job descriptions, text data must be transformed into numerical vectors , that's why we need vectorization , it is the process of converting text data into numerical representations (vectors) so it can be processed by machine learning or deep learning models. These vectors capture either word frequencies or contextual meaning, depending on the technique used.

5.3.1 TF-IDF (Term Frequency-Inverse Document Frequency)

TF-IDF is a traditional vector space model that converts text into sparse numerical vectors based on word frequency and rarity across documents. Cosine similarity is then applied to determine textual overlap.
— Term Frequency (TF): measures the frequency at which a specific word (or group of words) appears in a document.
— Inverse Document Frequency (IDF): involves finding or understanding the importance of a term in a collection of documents. This technique helps identify the rarity of a term or word across all documents.

5.3.2 Embeddings BERT

BERT embeddings are the foundational vector representations produced by the BERT model. They are dense, high-dimensional vectors that capture the **semantic meaning of words, phrases, or entire documents** in context. These embeddings form the core of the vectorization technique used in our AI-based resume matching module . In this AI system, both the **job description** and the **candidate's resume** are converted into sentence-level embeddings using the `SentenceTransformer.encode()` function. This abstracts away the internal BERT operations and returns a fixed-length vector for each input. These two vectors are then compared using cosine similarity to quantify how semantically close the two texts are.

5.3.3 Comparaison between the different vectorization techniques

Technique	Advantages	Limitations
TF-IDF	• Simple and fast to compute • Easy to interpret • Effective for basic keyword matching • Requires no pre-training	• No understanding of context • Cannot handle synonyms or paraphrases • Ignores word order • Produces sparse, high-dimensional vectors
BERT Embeddings	• Captures deep semantic meaning • Context-aware representations • Handles synonyms, paraphrasing, and grammar • Ideal for semantic similarity and ranking	• Computationally intensive • Slower inference time • Requires pre-trained models • Less interpretable vectors

Table 5.1: Advantages and Limitations of TF-IDF vs. BERT Embeddings

BERT embeddings were selected over traditional vectorization methods like TF-IDF because they provide a deeper, contextual understanding of language. Unlike TF-IDF, which only considers word frequency, BERT captures the meaning of entire sentences by generating context-aware embeddings. This makes it far more effective for matching resumes to job descriptions, where similar concepts can be expressed in varied language. By leveraging BERT, the system is able to compare documents based on **semantic similarity**, resulting in more accurate and intelligent applicant ranking.

5.4 AI System Overview

The resume-job matching module integrated into this recruitment platform leverages natural language processing (NLP) to automatically evaluate how well a job seeker's resume aligns with a given job offer. This evaluation is primarily based on semantic similarity between the resume and job description, and the overlap of extracted technical skills. It aims to assist recruiters in identifying top candidates efficiently .

5.4.1 System functionality

The resume–job matching module is designed to assist recruiters by automatically evaluating and ranking job applicants based on how well their resumes align with a specific job posting. This functionality is centered on a dual-layered approach: **semantic similarity** and **skill-based comparison**, ensuring a balance between contextual relevance and technical qualifications.

1. **Step 1: Resume and Job Description Preprocessing** When a job seeker applies for a job, their resume (in PDF format) is uploaded and processed to extract its textual content. Similarly, the job description is already available as a structured text field in the system.

    - The extracted resume text is cleaned using natural language preprocessing (lowercasing, stopword removal, punctuation stripping, and lemmatization).
    - The job description undergoes the same preprocessing to ensure uniformity and consistency in comparison.

2. **Step 2 : Semantic Similarity Computation (BERT)** : To evaluate contextual relevance between the job description and the candidate's resume, the system uses BERT (Bidirectional Encoder Representations from Transformers), a pre-trained language model that understands sentence-level meaning.

    - Both the cleaned resume and job description are encoded into dense vector representations using the `all-MiniLM-L6-v2` model from `sentence-transformers`.
    - The system computes the **cosine similarity** between the two vectors to determine how semantically close the candidate's experience and the job requirements are.

    This score reflects the **contextual alignment** of the candidate with the role, even when exact keywords are not shared.

3. **Step 3: Skill Extraction and Matching :**
    In parallel, the system performs explicit skill matching between the resume and the job description:

    - A curated list of recognized IT and tech-related skills is used as a reference (`SKILLS_LIST`).
    - The system extracts relevant skills from both documents using pattern matching and NLP techniques.
    - It then compares the extracted skills from the resume against the job's required skills to compute:

  - The number of **matched skills**.
  - The list of **missing skills**.
  - A **skill score** based on the percentage of required skills found in the resume.

   This allows recruiters to assess technical fit objectively.

4. **Step 4: Scoring and Ranking** The system combines the semantic similarity score and the skill score into a **final weighted score**, typically calculated as:

$$\text{Final Score} = 0.8 \times \text{Skill Score} + 0.2 \times \text{Semantic Similarity Score}$$

   This final score is used to:

   - Rank all applicants for a given job.
   - Display the top-matching candidates directly to the recruiter.
   - Reduce manual screening by focusing attention on the most relevant profiles first.

5.4.2 Output example :

```
Cleaned Job Description: seek experienced senior network engineer lead infrastructure project global oper
ation ideal candidate strong background cisco juniper system network security voip hybrid cloud environmen
t role involve lead cross functional team optimize network performance implement disaster recovery strateg
y
Cleaned Resume: samuel moore
network engineer | system optimization | network security
emaillinkedin.comphiladelphia pennsylvania
summary
network engineer year experience specialize large scale
network optimization cybersecurity measure expert cisco juniper
network solution achieve reduction downtime strategic
upgrade configuration experience
senior network engineer
cisco system
 education
 master science network engineering
 stanford university
    stanford california
 bachelor science computer science
 university pennsylvania
    philadelphia pennsylvania
 language
 english
 native
 spanish
Extracted Job Skills: ['network security', 'juniper', 'leadership', 'cisco']
Extracted Resume Skills: ['juniper', 'network security', 'cisco']
Matched: ['juniper', 'network security', 'cisco']
Missing: ['leadership']
Semantic Score: 66.31%
Skill Score: 75.0%
Final Match Score: 73.26%
127.0.0.1 - - [24/May/2025 02:17:07] "POST /match HTTP/1.1" 200 -
```

Figure 5.3: BERT based matching output.

Figure 5.3 presents the output of an AI-based skill matching system. It shows the cleaned job description and resume, the extracted technical skills from each, and a comparison of matched and missing skills. The system calculates a semantic similarity score (66.31%), a skill match score (75.0%), and a final match score (73.26%) based on the overlap and relevance of skills between the resume and job description.

5.4.3 Benefits of the resume-job matching module :

The implementation of this matching module brings significant benefits to both sides of the recruitment process recruiters and applicants by automating and enhancing the evaluation of candidates based on relevance, not just keywords.

- **For Recruiters :**

1. **Reduced Time-to-Hire** : The system eliminates the need to manually scan every resume. Recruiters can immediately focus on top-ranked candidates based on objective scores.
2. **Enhanced Candidate Experience** By shortening delays in screening, recruiters can respond to candidates faster, improving the company's brand perception.
3. **Data-Driven Decision Making :** Each application is scored using a consistent methodology, allowing recruiters to justify shortlisting and hiring choices with transparent metrics.

- **For Job Seekers:**

1. **Increased Fairness and Visibility** : All candidates are evaluated using the same criteria, giving qualified individuals a fair chance regardless of formatting or writing style.
2. **Minimizes Human Bias** The AI-powered evaluation is consistent and free from unconscious biases often found in manual screening (e.g., name, school, gender).
3. **Less Risk of Being Overlooked :** With traditional manual screening, strong candidates might be missed. This system ensures applicants with relevant skills are ranked and seen.

5.5 BERT based semantic comparaison :

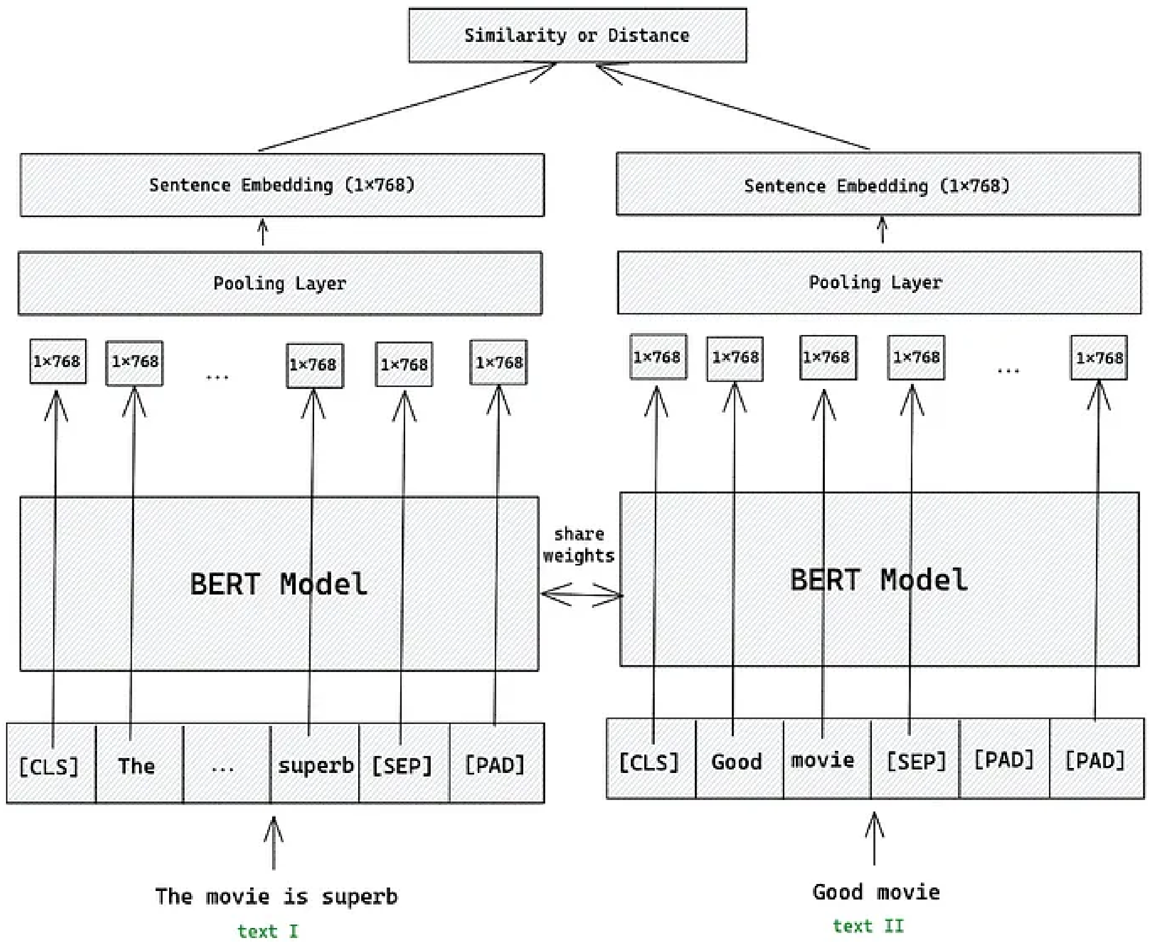

Figure 5.4: Semantic Matching Workflow Using BERT Sentence Embeddings
[25]

Figure 5.3 illustrates how semantic similarity is computed between two texts — in this case, a job description and a candidate's resume — using BERT-based sentence embeddings. This exact architecture underlies the AI component of the developed recruitment system.

Each input sentence (e.g., *"The movie is superb"* and *"Good movie"*, standing in for a job description and resume) is tokenized using the BERT tokenizer. This process breaks the input into subword units and adds special tokens:

- `[CLS]` marks the beginning of the sentence and is often used for classification tasks.
- `[SEP]` separates sequences or marks the end.
- `[PAD]` is added to ensure uniform input length for batch processing.

The tokenized inputs are passed independently into a pretrained BERT model (`all-MiniLM-L6-v2`) with shared weights, meaning both use the exact same pretrained parameters. Each token is transformed into a **768-dimensional contextual vector**, capturing the meaning of each word based on its surrounding context.

These token-level vectors are then processed by a **pooling layer**. In this system, **mean pooling** is applied — averaging all token embeddings (excluding `[PAD]`) to form a single vector that represents the entire sentence. This results in a fixed-size **sentence embedding** for each input.

Finally, the two sentence embeddings are compared using **cosine similarity**, producing a **semantic match score** between 0 and 1. This score reflects how similar the resume is to the job description in meaning — even if they use different words. In the recruitment system, this semantic score is combined with skill-based analysis to produce a final applicant ranking.

5.6 Technologies and used Tools

Figure 5.5: Python - Logo

A widely-used programming language known for its simplicity and strong support for AI and data science tasks [26]

Figure 5.6: Flask - Logo

A lightweight and flexible Python micro-framework used to create the RESTful API that handles communication between the frontend and the AI model. [27]

Figure 5.7: Spacy - Logo

An advanced NLP library used for preprocessing resumes and job descriptions. It performs tokenization, lemmatization, and stopword removal efficiently [28]

5.7 User interfaces for AI integration in the recruitment system

This section presents the user interfaces that integrate the AI-based applicant ranking module. These views allow recruiters to visualize match scores, access detailed evaluation metrics, and streamline the selection process based on AI-generated insights.

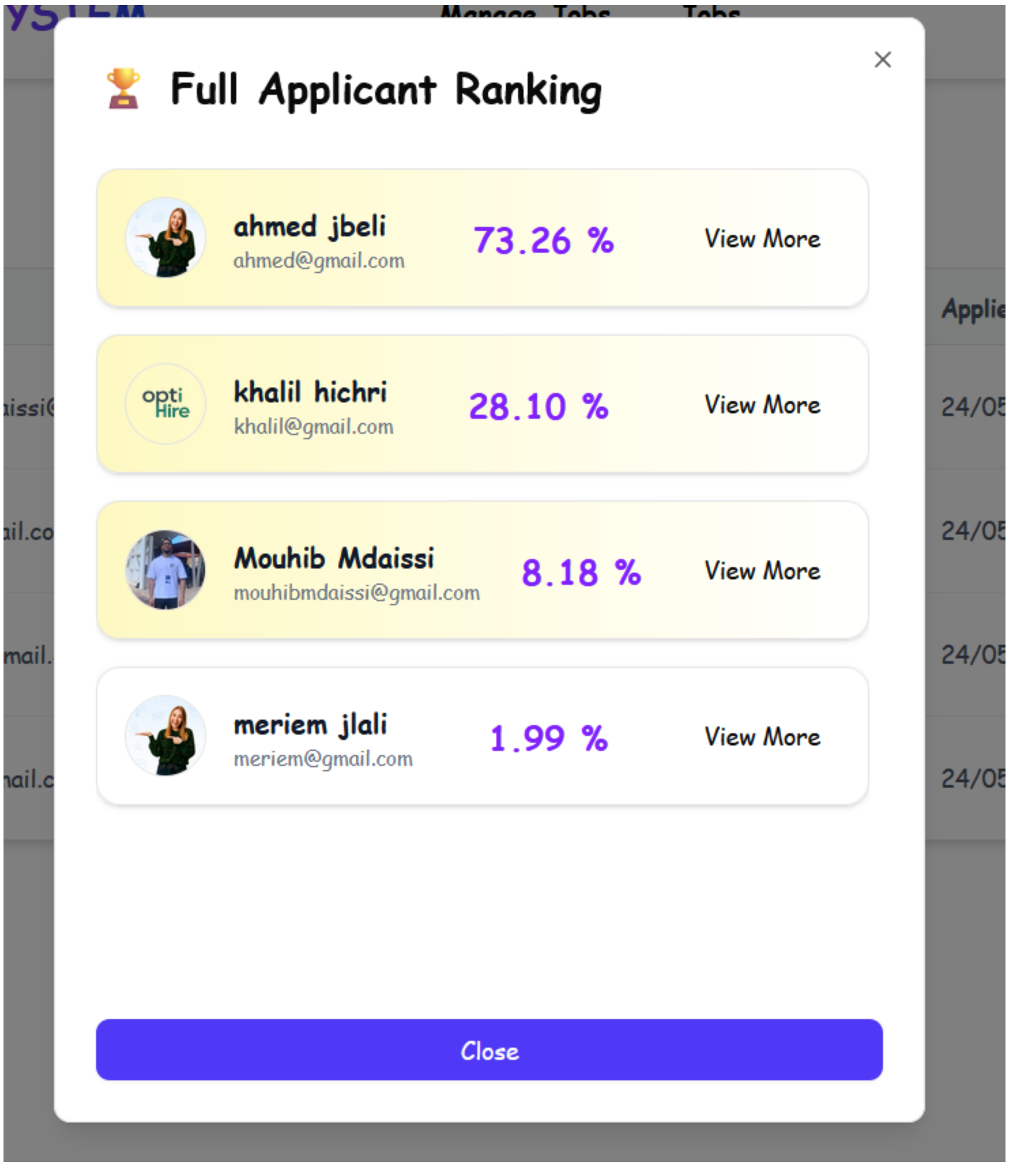

Figure 5.8: Applicant ranking modal

Figure 5.8 displays the ranked list of applicants based on their match scores. Recruiters can quickly view each candidate's score and access detailed profile or evaluation information via the "View Profile" and "View More" buttons.

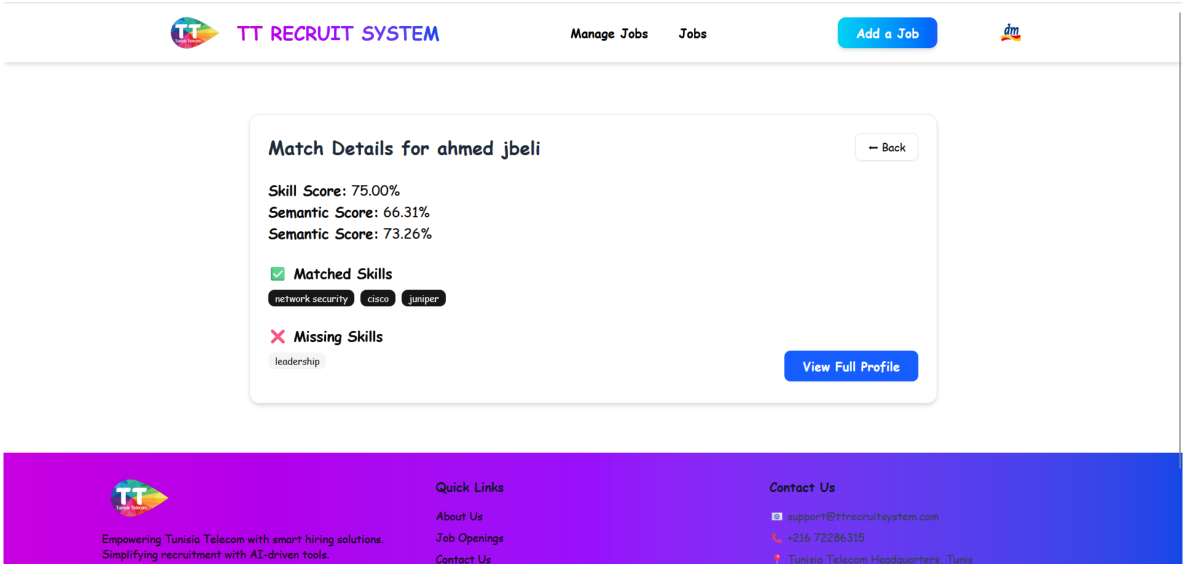

Figure 5.9: Match score details page

Figure 5.9 provides a breakdown of the matching analysis for a selected applicant. It presents the skill alignment and semantic relevance between the job posting and the candidate's profile, helping recruiters understand the strengths and potential gaps of each applicant.

5.8 Conclusion :

This chapter detailed the design and integration of the AI-based resume matching module into the recruitment system. Through an exploration of key concepts such as semantic similarity, BERT embeddings, and cosine similarity, we established the technical foundation enabling automated and intelligent resume-job comparisons. We highlighted how the system extracts relevant features from job descriptions and candidate resumes, computes similarity scores, and ranks applicants accordingly streamlining the hiring process for recruiters. Interfaces such as the ranking modal and detailed match breakdown were also introduced, offering recruiters clear visibility into match quality and decision-making support.

General Conclusion

This final-year project aimed to design and develop an intelligent recruitment system named **TT Recruit System**, in response to the real needs of Tunisie Telecom. The main goal was to modernize and simplify the recruitment process, which was previously handled through fragmented tools and manual procedures. The proposed platform offers an integrated and automated approach to managing job offers, candidate applications, and the overall evaluation process.

The system was built using the **MERN stack** (MongoDB, Express.js, React.js, and Node.js), which provided a coherent and efficient environment for full-stack development. MongoDB was used for flexible and scalable data storage, Express.js and Node.js handled the backend logic and APIs, and React.js ensured a responsive and user-friendly interface. This architecture allowed for smooth interaction between components and efficient handling of user data and workflows.

A key innovation of the system lies in its integration of **artificial intelligence** for CV parsing and semantic matching. By analyzing the content of resumes and comparing it to job descriptions, the system helps recruiters to identify the most relevant candidates automatically, saving time and improving decision accuracy.

The platform also includes essential features such as secure user authentication, role-based interfaces for recruiters and candidates, and automated email notifications at each important stage of the application process. Special attention was given to usability and real-world recruitment constraints—for instance, allowing only one active application per candidate at a time to reflect actual hiring policies.

Beyond its technical implementation, this project allowed us to strengthen our understanding of software architecture, web technologies, and real-world challenges in recruitment. It also offered an opportunity to apply artificial intelligence in a practical and impactful way.

In the future, the system could be enhanced by adding advanced features such as analytics dashboards for recruiters, integration of video interviews, the automatic generation of job offers based on the analysis of submitted resumes, thereby shifting from static job listings to a more dynamic, candidate-driven recruitment process..These improvements would make the platform even more comprehensive and better suited to large-scale recruitment needs.

In summary, **TT Recruit System** represents a modern, intelligent, and scalable recruitment platform that responds effectively to the evolving needs of Tunisie Telecom and sets the foundation for future growth and innovation.

Bibliography

[1] Wikipedia. Tunisietelecom. https://www.tunisietelecom.tn/particulier/, https://www.esri.com/en-us/lg/industry/telecommunications/stories/tunisie-telecom-case-study. Accessed April 2025.

[2] linkedin. Linkedin. https://www.linkedin.com/help/recruiter/answer/a6895082. Accessed April 2025.

[3] smartRecruiters. smartrecruiters. https://www.smbguide.com/review/smartrecruiters. Accessed April 2025.

[4] Glassador. Glassador. https://www.glassdoor.com/about. Accessed April 2025.

[5] Agile. Agile. https://www.atlassian.com/agile. Accessed April 2025.

[6] Scrum Alliance – What is Scrum? Scrum. https://www.scrumalliance.org/about-scrum. AccessedApril 2025.

[7] Grady Booch, James Rumbaugh, and Ivar Jacobson. *The Unified Modeling Language User Guide*. Addison-Wesley, 2nd edition, 2005.

[8] Microsoft. Visual studio code. https://code.visualstudio.com/, 2025. Accessed May 2025.

[9] Postman Inc. Postman: The collaboration platform for api development. https://www.postman.com. visited on May 2025.

[10] Git - distributed version control system. https://git-scm.com/. Accessed May 2025.

[11] Kazem Mammadov. The ultimate guide to github: Everything you need to know. https://kazemmdev.medium.com/the-ultimate-guide-to-github-everything-you-need-to-know-310b6173abb2, 2021. Accessed May 2025.

[12] MongoDB, Inc. Mongodb compass. https://www.mongodb.com/products/compass, 2025. Accessed May 2025.

[13] Meta Platforms, Inc. React – a javascript library for building user interfaces. https://reactjs.org/, 2025. Accessed May 2025.

[14] ECMA International. Javascript – ecmascript language specification. https://developer.mozilla.org/en-US/docs/Web/JavaScript, 2025. Accessed May 2025.

[15] Adam Wathan et al. Tailwind css – a utility-first css framework. https://tailwindcss.com/, 2025. Accessed May 2025.

[16] OpenJS Foundation. Node.js – javascript runtime. https://nodejs.org/, 2025. Accessed May 2025.

[17] OpenJS Foundation. Express – fast, unopinionated, minimalist web framework for node.js. https://expressjs.com/, 2025. Accessed May 2025.

[18] MongoDB Inc. Mongodb nosql database. https://www.mongodb.com. visited on May 2025.

[19] Andris Reinman. Nodemailer – send e-mails with node.js. https://nodemailer.com/, 2025. Accessed May 2025.

[20] Kenton Lee Kristina Toutanova Jacob Devlin, Ming-Wei Chang. Bert: Pre-training of deep bidirectional transformers for language understanding. https://arxiv.org/abs/1810.04805. Accessed May 2025.

[21] Greg Corrado Jeffrey Dean Tomas Mikolov, Kai Chen. Efficient estimation of word representations in vector space. https://arxiv.org/abs/1301.3781. Accessed May 2025.

[22] Mohamad Merchant. Semantic similarity with bert. https://keras.io/examples/nlp/semantic_similarity_with_bert/. Accessed: April 2025.

[23] Amit Singhal. Modern information retrieval: A brief overview. `https://www.sciencedirect.com/topics/computer-science/cosine-similarity`. Accessed May 2025.

[24] Israa Hamdine. Fine-tuning bert for semantic textual similarity with transformers in python. `https://thepythoncode.com/article/finetune-bert-for-semantic-textual-similarity-in-python`. Accessed: April 2025.

[25] Mustafa77. Semantic textual similarity using bert. `https://medium.com/@Mustafa77/semantic-textual-similarity-with-bert-e10355ed6afa`. Accessed: April 2025.

[26] Python Software Foundation. Python programming language. `https://www.python.org`. visited on May 2025.

[27] Pallets Projects. Flask - web development, one drop at a time. `https://flask.palletsprojects.com`. visited on May 2025.

[28] Explosion AI. spacy · industrial-strength natural language processing. `https://spacy.io`. visited on May 2025.

Summary :

TT Recruit System is an intelligent web-based platform developed to digitize and improve the recruitment process at Tunisie Telecom. It allows job seekers to apply for positions easily, while recruiters can post job offers, monitor applications, and benefit from AI-powered candidate ranking. With an intuitive interface and automated features, the platform enhances recruitment speed, efficiency, and decision quality. It supports the company's digital transformation by offering a centralized, scalable, and modern recruitment solution tailored to the needs of a major enterprise.

Keywords : Recruitment - Artificial Intelligence - Digitalization -Matching - Recruiters - Job seekers - Ranking - Applications - Resume .

Résumé :

TT Recruit System est une plateforme web intelligente conçue pour digitaliser et optimiser le processus de recrutement au sein de Tunisie Telecom. Elle permet aux candidats de postuler facilement à des offres d'emploi, tandis que les recruteurs peuvent publier des annonces, suivre les candidatures et bénéficier d'un système de tri basé sur l'intelligence artificielle. Grâce à une interface intuitive et des fonctionnalités automatisées, la plateforme améliore la rapidité, l'efficacité et la qualité des décisions de recrutement. Elle s'inscrit dans une démarche de transformation numérique, en offrant un outil moderne, centralisé et évolutif, adapté aux besoins d'un grand opérateur.

Mots Clés : Recrutement - Intelligence Artificielle - Digitalisation - Candidats - Recruteurs - Classement - Candidatures - CV .

الملخص:

هذا المشروع هو نظام توظيف ذكي قائم على الويب، تم تطويره بهدف رقمنة وتحسين عملية التوظيف داخل شركة اتصالات تونس. يتيح للمرشحين التقديم بسهولة على الوظائف، بينما يمكن لفريق التوظيف نشر العروض وتتبع الطلبات والاستفادة من نظام تصنيف قائم على الذكاء الاصطناعي. بفضل واجهته البسيطة وميزاته المؤتمتة، يعزز النظام سرعة وكفاءة وجودة قرارات التوظيف. ويُعد جزءًا من جهود التحول الرقمي من خلال توفير حل موحد ومرن وحديث يتماشى مع متطلبات مؤسسة كبيرة.

الكلمات المفتاحية: التوظيف – الذكاء الاصطناعي – الرقمنة – المطابقة – فريق التوظيف – طالبي الوظائف – التصنيف – الطلبات – السيرة الذاتية.